A correlated prior on galaxy bias across thin redshift bins

November 2026

1 Summary

The problem. The forecast measures DESI galaxy clustering alone: the angular band powers C_ℓ^{gg} of thin redshift bins, computed without the Limber approximation, with the Fourier scale cut at $k_{\max} = 0.075\,h/\mathrm{Mpc}$ and a linear bias model. The bins belong to six tracer groups, BGS, LRG1, LRG2, LRG3, ELG1 and ELG2, and four bin layouts are compared: bins of equal comoving width $\Delta d_{\mathrm{com}} = 62$ Mpc, and bins of fixed redshift width $\Delta z = 0.02$, 0.05 and 0.2. The figure of merit is $\mathrm{FoM}(w_0, w_a) = [\det \mathrm{Cov}(w_0, w_a)]^{-1/2}$, the inverse area of the error ellipse of the dark-energy equation-of-state parameters w_0 and w_a.

The default model ties all bins of a group to one amplitude: $b_i = B_g/D(z_i)$, where b_i is the linear bias of bin i, $D(z)$ is the linear growth factor normalized to $D(0) = 1$, and B_g is one free amplitude per group g. This tying is what makes thin bins powerful, and it fails in both directions at once. It is too rigid: known physics moves each bin away from the group trend at the percent level. And it is inaccurate: a drift the model cannot express does not average away, it displaces the fitted cosmology. Fitting data that carry the gentlest drift physics supplies, the slow debiasing of a conserved galaxy population, with the tied model shifts (w_0, w_a) by 11 to 21 times the model's own error ellipse, depending on the layout (Appendix F). The opposite extreme, one free bias per bin, is accurate by construction but spends part of the data measuring 76 biases.

The model. Every intermediate case in this note is one model. The bias of bin i in group g is

$$b_{1,i} \;=\; \frac{B_g}{D(z_i)}\,(1+\delta_i), \qquad \delta_i \;=\; s_g\,(z_i - \bar{z}_g) + \varepsilon_i, \tag{1}$$

with a correlated Gaussian prior on the residuals within a group,

$$\langle \varepsilon_i\,\varepsilon_j \rangle \;=\; \sigma_g^2\,\exp\!\left(-\frac{|z_i - z_j|}{L}\right), \tag{2}$$

and zero across groups. Here z_i is the bin redshift, $\bar{z}_g$ is the mean redshift of the group's bins, B_g and s_g are a free amplitude and a free tilt per group (flat priors), δ_i is the fractional deviation from the group trend, ε_i is the residual around the group's straight line, σ_g is the prior width of one bin's residual in group g, and L is the correlation length in redshift: bins much closer than L share their residual, bins much farther apart do not. The shape B_g/D is the constant-clustering-amplitude trend measured for DESI-like luminous red galaxies [6]; it is not the evolution of a conserved population, whose bias obeys $b - 1 \propto 1/D$ [1], and that difference is one reason the tilt must be free. In the Fisher matrix the prior becomes one projected precision block per group that charges only the residual shape, never the level or the line; Appendices H–L derive the projection.

The ladder. The paper reports four nested assumptions, from rigid to free.

- **Tied**: $b_i = B_g/D(z_i)$, one free amplitude per group and nothing else. The upper limit of the FoM.
- **Baseline**: the model of Eq. (1) with $\sigma_g = 3\%$ for BGS, $\sigma_g = 1\%$ for the other five groups, and $L = 0.2$. Each number has a stated physical source. The BGS width is sized to the curvature of its flux-limit drift: the $r < 19.5$ magnitude limit turns into a luminosity threshold that rises with redshift and drags the bias up by about 22% across the group, and a straight line removes all of that except a 2.4% rms curvature, which is what σ_g must hold (Appendix N). The 1% of the other groups is a stated hedge for selection curvature not modeled in this note, about ten times the residual a conserved population leaves after its line (rms at most 0.16% across the groups, Appendix E.4). $L = 0.2$ is comparable to the group spans, so the residual is smooth across a group.
- **Pessimistic**: the same model with $\sigma_g = 3\%$ in every group and $L = 0.05$: local selection features in every group, as large as the largest known curvature, on the scale over which the 4000 Å break crosses a filter edge ($\Delta z \approx 0.03$–0.05). The selection discussion of Appendix C.4 finds that mechanism but no measurement of its amplitude, so this rung grants it the largest measured size everywhere.
- **Independent**: one free bias per bin, flat priors. The lower limit.

Why a ladder and not one prior. A prior is plausible if the bias histories that physics gives us look like ordinary draws from it. That test is necessary, not sufficient, and it needs trusted histories; this note has a few derived drift laws and no mock catalogs. So the paper does not defend a single choice. It reports a reasonable case and a pessimistic case, and shows how the figure of merit depends on the assumption, inside two absolute limits. A referee who rejects the baseline moves down one rung; nothing needs retracting.

The cost of freedom. Figure 1 and Table 1 give the FoM of every rung on every layout. On the thin layouts the baseline keeps roughly half of the tied FoM and stays above the LSST Y1 3x2pt reference of 17.26 on the $\Delta d_{\rm com} = 62$ Mpc layout. The pessimistic prior costs another third to half. On $\Delta z = 0.2$ every group holds one or two bins, so the free amplitude and tilt already free every bin and the three lower rungs coincide. The absolute values inherit a known mismatch between the data vector's biases and the covariance's, which inflates every layout's FoM by a factor of about 2.4–3.2 (Appendix D); the comparison between rungs on a fixed layout holds the data vector and covariance fixed and changes only the prior.

The accuracy test. Each rung is tested against truths it may not be able to express. A truth sets every bin's bias to its group's known drift (the $r < 19.5$ flux limit in BGS, a conserved population elsewhere) plus a residual drawn from the baseline or the pessimistic prior; every rung fits 2000 truths per case with its bias prior centered on the tied shape. Figure 2 draws the figure of bias Δ_w, the distance of the fitted (w_0, w_a) from the truth in units of the rung's own ellipse, one panel per truth and one column per rung. A point above the dashed line $\Delta_w = 1/3$ is a biased fit, except in the column labeled "fit prior = truth scatter", whose ellipse already covers that shift. The verdict is the error inflation $\sqrt{\mathrm{E}[\chi^2_w]/2}$, where χ^2_w is Δ^2_w with the data

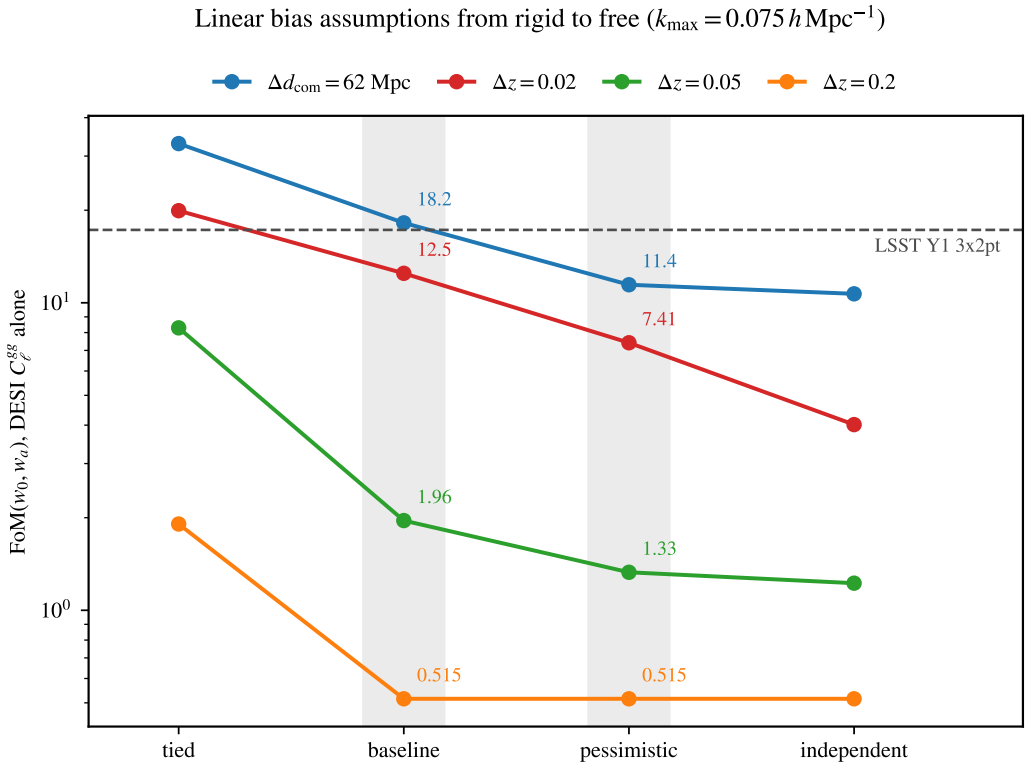


Figure 1: FoM(w_0, w_a) of DESI C_ℓ^{gg} alone at $k_{\max} = 0.075\,h/\mathrm{Mpc}$ on the four rungs of the linear-bias ladder, one line per bin layout. $k_{\max}$ is quoted at $z = 0$: each bin is cut by the ultraviolet-fraction criterion at $k_{\max}/D(\bar{z}_i)$ ($\alpha = 0.05$, $\bar{z}_i$ the bin's mean redshift) within the shipped mask. The rungs: tied, $b_i = B_g/D(z_i)$ with one free amplitude per group; baseline, the model of Eq. (1), a free amplitude and tilt per group plus a correlated residual with $\sigma_g = 3\%$ for BGS, 1% for the other groups and $L = 0.2$; pessimistic, the same model with $\sigma_g = 3\%$ in every group and $L = 0.05$; independent, one free bias per bin. The dashed line is the LSST Y1 3x2pt reference at 17.26. On $\Delta z = 0.2$ the free amplitude and tilt already free every one- and two-bin group, so the three lower rungs coincide at 0.515.

Layout	tied	baseline	pessimistic	independent
$\Delta d_{\rm com} = 62$ Mpc	32.94	18.19	11.45	10.69
$\Delta z = 0.02$	19.93	12.47	7.41	4.02
$\Delta z = 0.05$	8.29	1.96	1.33	1.23
$\Delta z = 0.2$	1.91	0.515	0.515	0.515

Table 1: FoM(w_0, w_a) of DESI C_ℓ^{gg} alone at $k_{\rm max} = 0.075\,h$/Mpc on the four rungs of the ladder, per layout. $k_{\rm max}$ is quoted at $z = 0$: each bin is cut by the ultraviolet-fraction criterion at $k_{\rm max}/D(\bar{z}_i)$ ($\alpha = 0.05$) within the shipped mask. The LSST Y1 3x2pt reference is 17.26.

noise added and E averages over truths and noise: it is 1 when the ellipse covers the truth as it claims, and above 1 it is the factor by which the ellipse is too small. The tied model is wrong under either truth: its ellipse is 8.5–18 times too small. The baseline is correct (1.00) when nature's local bias features stay within its own prior, but its ellipse is 1.5–2.3 times too small if they reach the pessimistic size; the baseline FoM rests on that assumption, and the pessimistic FoM is the safe statement. The pessimistic rung is correct (1.00) or conservative (0.87–0.99) under both truths. Appendix F derives both statistics. Scored for the matter density Ω_m and the primordial amplitude A_s, the truths give the same verdict (Appendix F.6).

What the two priors look like. Figures 3 and 4 draw both priors next to the known drifts. Under the baseline the neighbor residual rms at $\Delta z = 0.02$ is 0.44% (1.31% for BGS), the draws are smooth, and the known drifts look like ordinary members of the family; the curvature the BGS line leaves costs a χ^2 per degree of freedom of about 0.8 under the residual prior on the $\Delta d_{\rm com} = 62$ Mpc bins. Under the pessimistic prior the neighbor rms is 2.44% in every group: the histories keep their lines but gain local features nobody has measured, which is the point of that rung.

Higher-order bias. The companion note on higher-order bias builds the quadratic bias $b_{2,i}$ and the tidal bias $b_{s^2,i}$ of each bin on the per-bin linear biases of Eq. (1), through calibrated relations plus one free offset per group, so the higher-order sector drifts with the linear one. At $k_{\rm max} = 0.075\,h$/Mpc that sector adds the parameters of the two-page model summary: the offsets $\Delta_{2,g}$ and $\Delta_{s,g}$, one counterterm amplitude α_0 and one shot-noise deviation SN_0 per group, each with its stated Gaussian prior.

Limits. The ladder assumes linear bias. A linear-bias fit to a one-loop truth at the same cut is already biased: the truth is model A of `notes/higher_order_bias`, with the quadratic bias b_2 on its calibration $b_2(b_1)$ and the tidal bias b_{s^2} on co-evolution. On the $\Delta d_{\rm com} = 62$ Mpc layout the fit moves Ω_m by 2.0–2.4σ, (w_0, w_a) by 0.7–1.1σ and A_s by 0.9–1.2σ on every rung, in units of the rung's own marginalized errors. The rerun with the exact projection of the one-loop white-noise constant reproduces these numbers, and the companion note `notes/linear_bias_kmax` gives the full scan over $k_{\rm max} = 0.05$–$0.25\,h$/Mpc. Only $k_{\rm max} = 0.05\,h$/Mpc keeps every shift below $\sigma/3$, and only on the $\Delta d_{\rm com} = 62$ Mpc layout on every rung; on $\Delta z = 0.05$ it passes only for the tied and baseline rungs and only for the truth on the calibrations; on $\Delta z = 0.02$ that cut keeps no band at all, and on $\Delta z = 0.2$ it already fails. The forecast with the one-loop model is `notes/correlated_bias_prior_v2` (this note under model A). This note's FoMs are the linear-bias reference, not a final forecast.

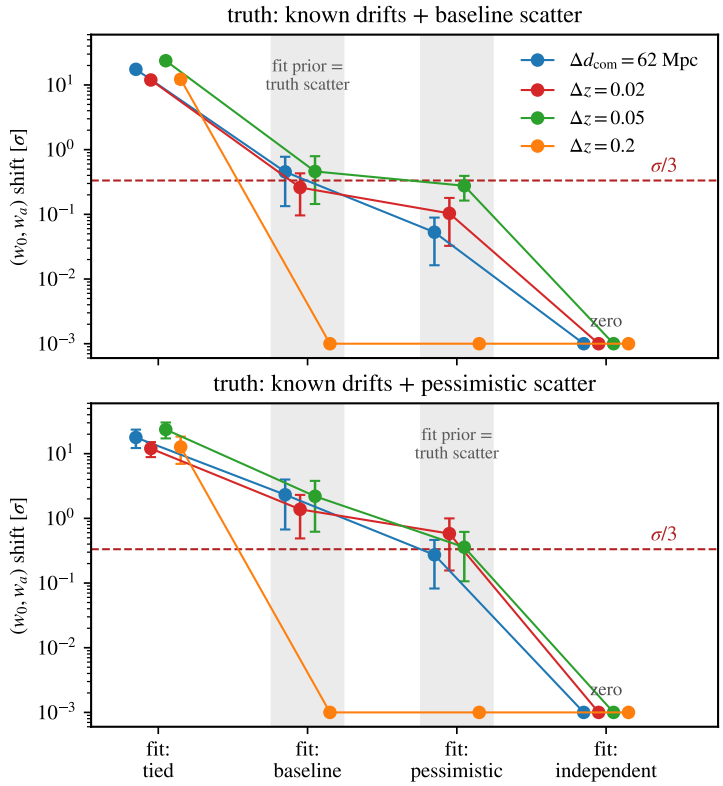


Figure 2: The accuracy test with random truths. In each panel the true bias of bin i is $b_i = B_g\,(1 + \delta_i^{\rm drift} + \varepsilon_i)/D(z_i)$: the drift is the $r < 19.5$ flux-limit deviation in BGS and the conserved-population deviation in the other five groups; the residual ε_i is Gaussian with covariance $\sigma_g^2\, e^{-|z_i - z_j|/L}$ inside a group, zero across groups, drawn from the baseline prior ($\sigma_g = 3\%$ for BGS, 1% otherwise, $L = 0.2$; top) or the pessimistic prior ($\sigma_g = 3\%$ everywhere, $L = 0.05$; bottom). The 2000 truths per panel are each fitted with every rung, every rung's bias prior centered on the tied shape $\delta = 0$: the baseline is here a model under test, not the truth. Points: the mean figure of bias Δ_w, the fitted (w_0, w_a) shift in units of that rung's own error ellipse; bars: the 16th–84th percentiles; colors: the four bin layouts; dashed line: the $\sigma/3$ threshold. The axis is logarithmic; shifts below 10^{-3} are drawn at 10^{-3} and labeled "zero": the independent rung absorbs any truth exactly, and on $\Delta z = 0.2$ every group holds one or two bins, so the free amplitude and tilt do the same for every rung below tied. The gray columns mark the baseline and the pessimistic rung; "fit prior = truth scatter" marks the rung whose prior equals the truth's residual distribution. There the mean shift is not a bias, since that rung's ellipse already covers exactly this scatter; its error inflation (Table 8) is 1.00. Reading the panels: the tied model fails both truths by more than an order of magnitude; the baseline absorbs its own scatter (top) but fails the pessimistic truth (bottom), where its ellipse is 1.5–2.3 times too small; the pessimistic rung passes both.

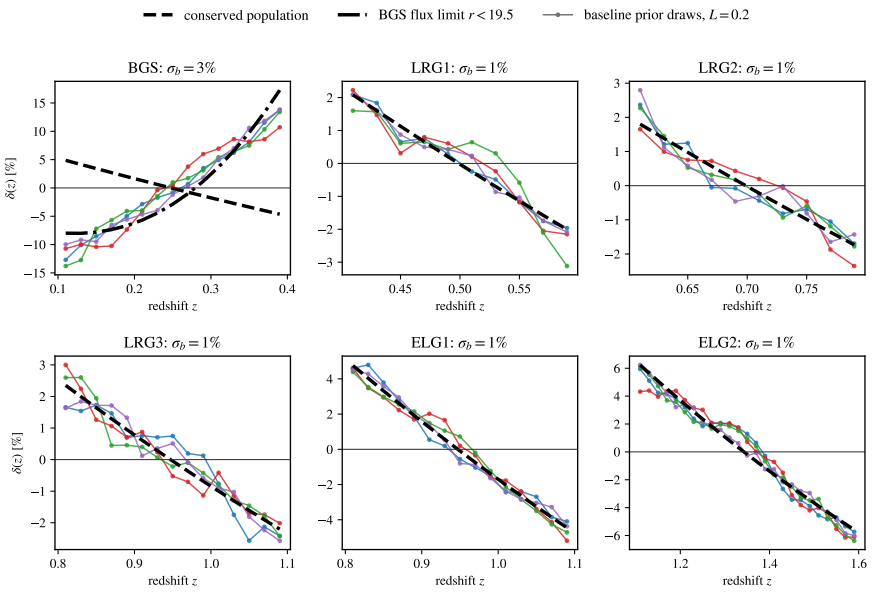


Figure 3: The baseline prior next to the known drifts, group by group, on the $\Delta z = 0.02$ bins. Thick dashed: the conserved-population drift. Thick dash-dot, on BGS: the $r < 19.5$ flux-limit drift. Thin curves with dots: four draws of the deviation $\delta(z)$ from the baseline prior (σ_g in each panel title, $L = 0.2$), each with its tilt set to the known drift's own straight line, so draw and drift share a line and differ by the residual ε. The neighbor residual rms is 0.44% at $\Delta z = 0.02$ (1.31% for BGS).

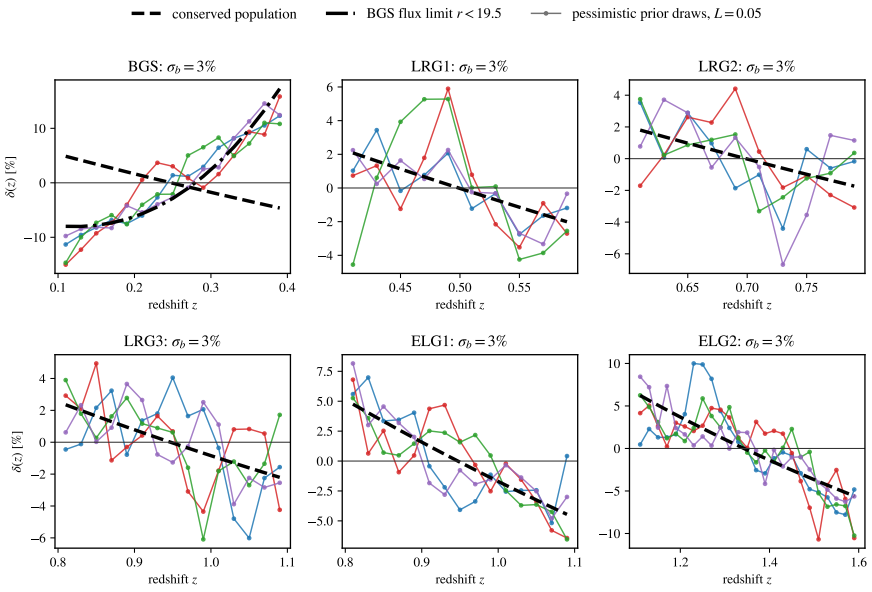


Figure 4: The pessimistic prior in the format of Figure 3: $\sigma_g = 3\%$ in every group and $L = 0.05$, drawn around the known drifts' lines. The neighbor residual rms is 2.44% at $\Delta z = 0.02$. The histories keep the drift lines but add local selection features of the size and length this rung assumes.

A Introduction: why an intermediate bias model

A Fisher forecast of galaxy clustering in thin redshift bins must decide how to parametrize the galaxy bias of each bin. The bins belong to a few broad groups: the DESI tracer samples and their chunks, such as BGS, LRG1, LRG2, LRG3, ELG1 and ELG2. Within one group the galaxies are selected the same way, so their biases are expected to be similar but not identical.

The default model ties all bins of a group to a single amplitude. If $D(z)$ is the linear growth factor, normalized to $D(0) = 1$, the tied model sets

$$b_i \;=\; \frac{B_g}{D(z_i)}, \tag{3}$$

where b_i is the linear bias of bin i, z_i is the bin's redshift, and B_g is one free amplitude per group g. The $1/D$ evolution keeps the clustering amplitude $b(z)\,D(z)$ of the group constant, the behavior measured for DESI-like luminous red galaxies [6]. It is not the law of a fixed set of galaxies that follow the flow of matter ("passive evolution"); Appendix C.1 derives that law and the small difference between the two.

The opposite extreme frees one bias per bin, with flat priors and no connection between neighbors ("independent"). The dark-energy figure of merit of the forecast depends on which extreme is chosen, and the question is how much of it rests on the tying itself. To answer that, this note builds the intermediate model: bins may deviate from the group's trend, but neighboring bins must deviate *together*.

B The models

B.1 Tied, independent, and correlated biases

The model multiplies the tied bias by a fractional deviation δ_i,

$$b_i \;=\; \frac{B_g}{D(z_i)}\,\big(1+\delta_i\big), \tag{4}$$

with B_g still free (flat prior). The deviations receive a Gaussian prior that is correlated inside a group,

$$\big\langle \delta_i\,\delta_j \big\rangle \;=\; \sigma_b^2\,\exp\!\left(-\,\frac{|z_i - z_j|}{L}\right), \tag{5}$$

and zero across groups. Here σ_b is the prior standard deviation of a single bin's fractional deviation, and L is the correlation length in redshift: bins much closer than L deviate almost as one, bins much farther apart deviate independently.

In the Fisher matrix the convenient parameters are the per-bin amplitudes

$$\beta_i \;\equiv\; B_g\,\big(1+\delta_i\big), \qquad \text{fiducial } \beta_i = B_g, \tag{6}$$

so that $b_i = \beta_i / D(z_i)$. Since $\beta_i - B_g = B_g\,\delta_i$, the prior covariance of the β_i within one group is

$$\mathsf{C} \;=\; s^2\,\mathsf{R}, \qquad s \;=\; \sigma_b\,B_g, \qquad \mathsf{R}_{ij} \;=\; \exp\!\left(-\,\frac{|z_i - z_j|}{L}\right). \tag{7}$$

The spread s carries the units of a bias amplitude; the correlation matrix R is dimensionless, with ones on the diagonal.

B.2 What σ_b and L mean

The kernel of Eq. (5) has two parameters, and they control two different things. The diagonal gives the wander of a single bin,

$$\mathrm{Var}(\delta_i) \;=\; \sigma_b^2, \tag{8}$$

so σ_b sets how far any one bin may drift from the group's trend $B_g/D(z)$. The correlation coefficient r of two bins i and j follows from the same kernel. By definition,

$$r \;=\; \frac{\langle \delta_i \delta_j \rangle}{\sqrt{\langle \delta_i^2 \rangle \, \langle \delta_j^2 \rangle}} \;=\; \frac{\sigma_b^2 \, e^{-\Delta z/L}}{\sigma_b^2} \;=\; e^{-\Delta z/L}, \tag{9}$$

with $\Delta z = |z_i - z_j|$ the bin separation. The amplitude σ_b cancels: the correlation is set by the separation and L alone.

The jump between two bins now follows from the two pieces above: with $\mathrm{Var}(\delta_i) = \sigma_b^2$ and $\mathrm{Cov}(\delta_i, \delta_j) = \sigma_b^2 \, e^{-\Delta z/L}$, the variance of the difference is $\mathrm{Var}(\delta_i) + \mathrm{Var}(\delta_j) - 2\,\mathrm{Cov}(\delta_i, \delta_j)$:

$$\mathrm{Var}(\delta_i - \delta_j) \;=\; 2\,\sigma_b^2 \left(1 - e^{-\Delta z/L}\right). \tag{10}$$

For separations much smaller than the correlation length, $\Delta z \ll L$, the exponential expands as $e^{-x} \approx 1 - x$, and

$$\mathrm{Var}(\delta_i - \delta_j) \;\approx\; \frac{2\,\sigma_b^2 \, \Delta z}{L}, \qquad \text{rms jump} \;\approx\; \sigma_b \sqrt{\frac{2\,\Delta z}{L}}. \tag{11}$$

So L sets the correlation between bins, σ_b sets the wander of each bin, and *together* they set the jump a thin bin may take relative to its neighbor.

The combination that controls small separations is the roughness

$$\kappa \;\equiv\; \frac{\sigma_b}{\sqrt{L}}, \qquad \mathrm{Var}(\delta_i - \delta_j) \;\approx\; 2\,\kappa^2 \, \Delta z. \tag{12}$$

The second form is the signature of diffusion: the variance of the increment grows linearly with the separation, exactly as the distance walked by a random walker grows linearly in time. On scales well below L the deviation curve $\delta(z)$ therefore looks like a random walk with diffusion coefficient κ^2; only on scales beyond L does the finite wander σ_b pull it back toward zero.

Worked example. Take neighboring thin bins with $\Delta z = 0.02$ and correlation length $L = 0.05$. Then $\Delta z/L = 0.4$ and

$$r \;=\; e^{-0.4} \;=\; 0.670. \tag{13}$$

The exact rms jump from Eq. (10) is $\sigma_b \sqrt{2\,(1 - 0.670)} = 0.81\,\sigma_b$: for $\sigma_b = 10\%$ the neighboring bins may differ by 8.1% (one sigma), and for $\sigma_b = 1\%$ by 0.81%. The Taylor form of Eq. (11) gives instead $\sigma_b \sqrt{0.8} \;=\; 0.89\,\sigma_b$, an 8.9% jump. The approximation overestimates because $\Delta z/L = 0.4$ is not actually small: the expansion replaces $1 - e^{-x}$ by x, and x exceeds $1 - e^{-x}$ already at second order ($x^2/2$), by about twenty percent here; the square root halves that, so the rms jump is overestimated by about ten percent.

B.3 The free group amplitude and the projection

A plain Gaussian prior with covariance C would penalize moving all β_i of a group up or down together, but that uniform shift is exactly the free amplitude B_g, which the tied model never constrains. Keeping the penalty would make every comparison against the tied model unfair. The repair removes the uniform direction from the prior: with $\mathbf{1} = (1, \ldots, 1)^\mathsf{T}$ the ones vector with n entries (n the number of bins in the group), the prior precision of a group's amplitudes is the projection

$$\mathsf{P} \;=\; \mathsf{C}^{-1} \;-\; \frac{\mathsf{C}^{-1}\mathbf{1}\,\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}}{\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\mathbf{1}}. \tag{14}$$

Four statements, proven in the appendices, carry the construction. $\mathsf{P}\,\mathbf{1} = 0$: a uniform shift of the whole group costs nothing, so B_g stays as free as in the tied model. A pure shape change keeps exactly the penalty the plain prior gives it. P is positive semidefinite with rank $n - 1$: of a group's n directions, one is free and the other $n - 1$ are charged. And P is precisely the prior obtained by sampling B_g with a flat prior and marginalizing it. Appendix H computes P for two bins, Appendix I derives these statements three ways, and Appendix J treats n bins, including the tied and independent limits.

Across groups the prior of Eq. (5) is zero, so the full prior precision is block diagonal: one projected block per group, each group with its own free amplitude. How the blocks enter the Fisher matrix is Appendix P.

C Which L and σ_b are physical

Which deviations from the tied trend does known galaxy-bias physics produce? One case of bias evolution can be solved in closed form: a population whose members are conserved. This section derives that case, measures its deviation from the tied trend, compares the deviation with what the prior permits, weighs the published measurements, and ends at the scanned grid and the range it covers.

C.1 A conserved population: the bias continuity equation

Take a sample whose membership never changes: no galaxy enters, none leaves. Assume the galaxies move with the matter on large scales (no velocity bias), and work in linear theory.

Under these assumptions the matter overdensity δ_m (the fractional fluctuation of the matter density) and the galaxy-count overdensity δ_g obey the same linear continuity equation, with the same peculiar velocity field on the right-hand side: number conservation gives each left-hand side, and the common motion gives the shared right-hand side. Subtracting the two equations, the excess clumping $\delta_g - \delta_m$ is frozen: whatever clumping difference the sample is born with never changes. Appendix M spells out every step.

Let z_c be a reference redshift where the sample has linear bias b_c (linear bias means $\delta_g = b\,\delta_m$), and let the matter grow as $\delta_m \propto D$. Carrying the frozen difference from z_c to any other redshift gives [1]

$$b(z) - 1 \;=\; \big(b_c - 1\big)\,\frac{D(z_c)}{D(z)}. \tag{15}$$

The excess clumping set when the galaxies formed stays fixed while the matter clustering grows underneath it, so the bias decays toward one with time. Tegmark and Peebles [2] extend the

calculation to stochastic bias, where the conserved galaxy field traces the matter imperfectly (correlation $\rho_{gm} < 1$), and find the same decay of b toward one; Appendix M summarizes their result.

The tied shape $b = B/D$ freezes a different quantity. Constant $b\,D$ means $\delta_g = b\,\delta_m \propto (B/D)\,D = B$: the galaxy clustering amplitude itself does not grow. A conserved population never does that; its δ_g keeps growing with the matter piece $\delta_m(z)$ (Appendix M). Constant $b\,D$ instead describes a selection: at each redshift it picks a different and rarer set of objects, whose stronger clumping compensates the smaller D.

C.2 The deviation from the tied trend

Place the reference at a group's midpoint, z_c, with $b_c = B/D(z_c)$ from the group's fiducial amplitude B. The tied bias is $b_{\rm tied}(z) = b_c\,D(z_c)/D(z)$ and the conserved bias is Eq. (15). Their ratio defines the deviation $\delta(z)$ of Eq. (4), $b = b_{\rm tied}(1+\delta)$:

$$1+\delta(z) \;=\; \frac{1+(b_c-1)\,D(z_c)/D(z)}{b_c\,D(z_c)/D(z)} \;=\; \frac{D(z)/D(z_c)+b_c-1}{b_c}, \qquad \delta(z) \;=\; \frac{D(z)/D(z_c)-1}{b_c}. \tag{16}$$

At the midpoint $\delta = 0$. At lower redshift the matter has grown (D larger than $D(z_c)$), so $\delta > 0$; at higher redshift $\delta < 0$. Because $D(z)$ is smooth, $\delta(z)$ is nearly linear in z across a group.

Table 2 evaluates Eq. (16) in flat ΛCDM with $\Omega_m = 0.3$, using the project's fiducial amplitudes. In the language of the prior, a conserved population is a smooth tilt: coherent over the whole group, so L longer than the group width, with an amplitude of a few percent at the group ends and a step of 0.3–0.7% between neighbors at $\Delta z = 0.02$.

Group	B	z range	b_c	δ at the group ends	per $\Delta z = 0.05$
BGS	1.34	0.1–0.4	1.5	+5.3%, −4.9%	1.7%
LRG1	1.70	0.4–0.6	2.2	+2.4%, −2.2%	1.1%
LRG3	1.70	0.8–1.1	2.7	+2.6%, −2.3%	0.8%

Table 2: The conserved-population deviation $\delta(z)$ of Eq. (16) from the tied trend, for flat ΛCDM with $\Omega_m = 0.3$, the reference z_c at the group midpoint, and $b_c = B/D(z_c)$ from the project fiducials. The last column is the change of δ across one $\Delta z = 0.05$ step at the midpoint.

Figure 5 draws the shapes behind the table. Look at any group: the dashed conserved curve is flatter than the solid B_g/D, because the frozen excess clumping is divided by a growing D, so the conserved bias rises more slowly toward high redshift. On BGS, the dash-dot flux-limit curve is the opposite: much steeper than B_g/D, and convex.

C.3 The prior's yardstick: the neighbor rms

The prior's quantity to compare with this tilt is the rms change between neighbors of Eq. (10). Table 3 evaluates $\sigma_b\sqrt{2\,(1-e^{-\Delta z/L})}$ at several (σ_b, L) pairs and for the conserved population of Table 2.

At $\sigma_b = 1\%$ with $L = 0.1$ the prior allows neighbor differences of 0.6% over $\Delta z = 0.02$, matching the conserved drift. At $\sigma_b = 3\%$ with $L = 0.05$ the allowance is several times that

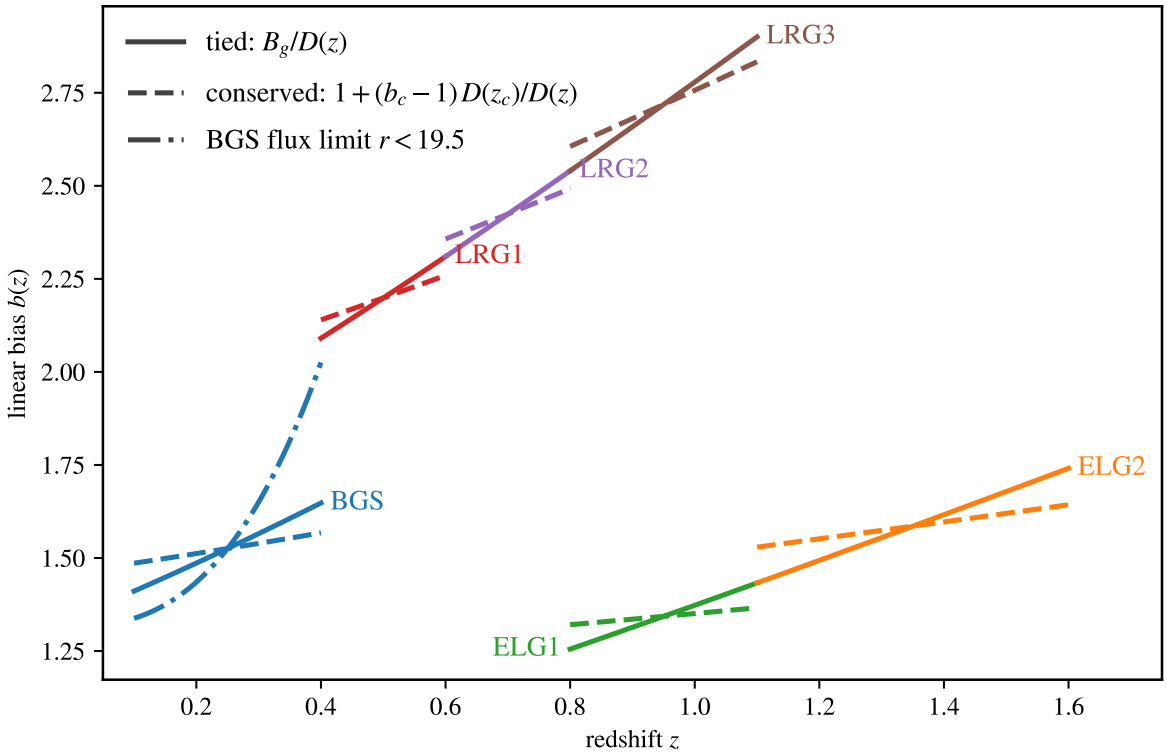


Figure 5: The bias $b(z)$ of every group. Solid: the tied shape $B_g/D(z)$. Dashed: a conserved population anchored at the group midpoint, $b(z) = 1 + (b_c - 1)\,D(z_c)/D(z)$ (Eq. (15)). Dash-dot, on BGS only: the $r < 19.5$ flux-limit bias, anchored at the same midpoint.

Case	over $\Delta z = 0.02$	over $\Delta z = 0.05$
$\sigma_b = 3\%$, $L = 0.05$	2.4%	3.4%
$\sigma_b = 1\%$, $L = 0.1$	0.6%	0.9%
$\sigma_b = 5\%$, $L = 0.02$	5.6%	6.8%
$\sigma_b = 10\%$, $L = 0.05$	8.1%	11.2%
$\sigma_b = 20\%$, $L = 0.02$	22.5%	27.1%
conserved population	0.3–0.7%	0.8–1.7%

Table 3: The rms change of the bias deviation between two bins, $\sigma_b\sqrt{2\,(1 - e^{-\Delta z/L})}$, at several (σ_b, L) pairs, and for the drift of a conserved population (the per-group range of Table 2).

drift, and at $\sigma_b = 10\%$ or 20% it reaches 8–27%, beyond any bias evolution identified in this note.

C.4 What breaks conservation in a selected sample

Four mechanisms break the conserved-population assumptions.

Threshold drift. The target selection holds a roughly constant comoving density until the flux limit bites; past that redshift the sample keeps only its brighter members, which sit in more massive halos and are more biased. The resulting bias change is smooth in redshift, with L comparable to the group width.

Color-selection features. The selection cuts in color space can respond quickly in redshift when a spectral feature crosses a filter edge. The sharpest common feature is the 4000 Å break: a filter edge 100–200 Å wide maps to $\Delta z \approx 0.03$–0.05 at that break, and photometric scatter smears the transition further. Its amplitude in bias is unknown.

Mergers. Two members that merge become one, so the number of galaxies is not conserved even before any selection acts. The size of this effect on these samples is unknown.

Emission-line selection. An ELG enters the sample while its emission lines are strong, so the membership follows the star-formation phase of each galaxy rather than one conserved set of objects.

How much each of these moves the bias, and over which Δz, is not known from theory alone; it is a question about the measured selection function of each sample.

C.5 What the measurements say

The mechanisms above can be weighed against published measurements. Each statement below carries a tag: *measured* means it is read from the cited source; *derived* means it is computed here from measured inputs; *assumed* means it is a judgment not yet backed by a measurement. Equation (15) itself is from Fry [1], which has no arXiv record; Tegmark and Peebles [2] (section "After the galaxy formation epoch") rederive it. Appendix O summarizes each paper below: the sample, the method and the numbers.

Massive red galaxies follow the conserved shape to the precision of the published tests. The passive-evolution analyses find merger corrections of a few percent per Gyr (measured [3, 4, 5]), and the DESI-like LRG bias of [6] stays within $\pm 2.3\%$ of the constant clustering amplitude $b = 1.5/D$ at a bin spacing of $\Delta z \approx 0.1$ below $z = 0.83$, with the highest bin $6.4 \pm 6.4\%$ high (measured biases; residuals derived here).

The selections drift. The LRG cuts hold the comoving density constant at $0.4 < z < 0.8$ and hand over to a fixed flux limit beyond (measured [7]); through the threshold-bias relation of [12] (measured), that flux limit makes the LRG3 bias drift by 6–11% per $\Delta z = 0.1$ on top of the B/D shape (derived), matching the low end of the measured highest-bin offset. The magnitude-limited BGS (measured limit, [15]) drifts by 10–25% across its group through the same mechanism (derived).

The ELG bias shape is stable: HOD fits in two bins covering $0.8 < z < 1.6$ agree within one sigma (measured [18]). The one short-separation mechanism found in any tracer is the set of redshift windows where the [OII] 3727 Å doublet, which carries the ELG redshift, lands on sky emission lines and loses signal-to-noise, especially at $z > 1.5$ (mechanism measured [17]): each window is $\Delta z \approx 0.001$–0.01 wide, and its amplitude in bias is assumed, of order 1–2%: a judgment from the weak dependence of ELG bias on [OII] flux, not a measured bound.

None of the measurements collected here resolves bias below $\Delta z \approx 0.1$. The finest binning is the $\Delta z \approx 0.1$ of [6]; [18] use 0.3 and 0.5; and the lensing tomography of the same samples [9, 10, 11], built on the photometric samples of [8], uses about 0.15 with one free bias per bin. Below $\Delta z \approx 0.1$ the thin-bin statements of this note rest on the continuity equation and on the selection arguments.

C.6 Drift versus loss of memory between bins

Every deviation found above, except the sky-line windows, is a monotonic drift: the conserved-population tilt, the LRG3 magnitude-limit drift and the BGS flux-limit drift each move in one direction across their group. A monotonic drift is a coherent, long-correlation deformation, a tilt; it is not bin-to-bin decorrelation, so it does not argue for a short L. Within the exponential kernel this note analyzes, the measured drifts point to the long end of the scanned L grid, and only the sky-line windows point to the short end.

C.7 The scan and the scanned range

The forecast scans

$$L \in \{0.02,\ 0.05,\ 0.1,\ 0.2\} \quad \text{(in redshift)}, \qquad \sigma_b \in \{1\%,\ 3\%,\ 5\%,\ 7\%,\ 10\%\}, \tag{17}$$

and the figures of Appendix D mark two of its elements: the kernel scan's reference point, $L = 0.05$ and $\sigma_b = 3\%$ (the diamond their legends label "baseline"), and the scanned range $L =$ 0.02–0.1, $\sigma_b = 1$–5% (the band their legends label "plausible"). Neither is this note's baseline. The stated assumptions of Section 1 set that, on the memory model, and the roughness analysis of Appendix G explains why a region in (L, σ_b) is the wrong object to promote: for smooth deviations the kernel constrains only the combination $\sigma_b/\sqrt{L}$. The reasoning below sized the range and stays as the record of that sizing.

The upper end of L rests on a physical argument: a sample's flux limit and color cuts change which galaxies enter the sample as redshift grows, and that change of galaxy mix happens over

$\Delta z \sim 0.1$, so bias deviations should not be assumed coherent over separations much beyond that: $L \lesssim 0.1$. (Appendix G weighs this argument against the degeneracy above.)

In Table 3, the reference point allows neighbor differences of 2.4% over $\Delta z = 0.02$, several times the conserved drift; the gentlest corner of the scanned range ($\sigma_b = 1\%$, $L = 0.1$) matches that drift, and the sharpest ($\sigma_b = 5\%$, $L = 0.02$) allows roughly eight times it. The rows $\sigma_b =$ 10% and 20% show what the conserved-population comparison rules out: neighbor differences of 8–27%. Both sit above the scanned range, and $\sigma_b = 20\%$ is outside the scanned grid.

Shape freedom costs dark-energy information: the Fisher matrix spends part of the data constraining the bin biases instead of w_0 and w_a; Appendix D measures the cost across the scanned grid. Relative to a conserved population, the figure of merit computed anywhere in the scanned range except its gentlest corner is pessimistic.

This comparison is not a proof, because a DESI sample is not conserved: the selection, not gravity, decides which galaxies enter each bin.

D Results

The forecast behind the scan is the FoM(w_0, w_a) of the DESI clustering band powers alone (C_ℓ^{gg}, no shear), computed non-Limber with the ultraviolet-fraction Fourier scale cut at $k_{\max} = 0.075\,h/\mathrm{Mpc}$ of [19], which follows DeRose and Chen [20]. The cut keeps a band while the ultraviolet fraction $|C_\ell^{\mathrm{UV}}|/(|C_\ell^{\mathrm{UV}}| + |C_\ell^{\mathrm{IR}}|)$ stays below $\alpha = 0.05$, where the two pieces are the band power computed from the power spectrum above and below the split $k_{\max}/D(z)$. The figure of merit is the usual inverse area of the (w_0, w_a) error ellipse, $\mathrm{FoM} = [\det \mathrm{Cov}(w_0, w_a)]^{-1/2}$: larger is better. Four thin-bin layouts are scanned: the $\Delta d_{\mathrm{com}} = 62$ Mpc layout (bins of equal comoving width) and the fixed-width layouts with $\Delta z = 0.02$, 0.05 and 0.2. In each layout the grid of Eq. (17) is bracketed by the tied and independent models.

Forecast setup. The survey covers 12,300 square degrees. The data vector is the project's measured DESI clustering block: the band powers of every bin auto-spectrum plus the measured adjacent cross-bin pairs (76 autos and 97 pairs on the $\Delta d_{\mathrm{com}} = 62$ Mpc layout), in 53 bands of width $\Delta\ell = 10$ over $\ell = 15$–544. The covariance is the source forecast's Gaussian matrix (cosmic variance plus shot noise), computed for 14,000 square degrees, with the inverse rescaled to the survey area. The model is linear bias with linear Kaiser redshift-space distortions; a per-group velocity dispersion smears each bin's $n(z)$ (the Fingers-of-God convolution); magnification is not modeled. The Fisher matrix varies seven cosmological parameters ($10^9 A_s$, n_s, H_0, Ω_b, Ω_m, w_0, w_a, all with flat priors, at the project examples' fiducial values) plus the per-bin biases and the per-group velocity dispersions, the latter with their Gaussian priors of widths 30–145 km/s. Derivatives are finite differences; on $\Delta z = 0.2$ they are recomputed at four step sizes and two CAMB accuracy settings, and the FoM agrees to 0.6% across the steps, the numerical side of the Fisher approximation's validity.

Preliminary: linear bias only. The forecast models the galaxy field with linear bias alone, with no magnification; the absolute FoM values will move when higher-order bias terms enter the model. A second caveat is larger. The DESI covariance files are computed with each group's raw bias (1.34, 1.70, 0.84) while the data vector uses $b_i = B_g/D(\bar{z}_i)$, which understates the high-redshift variances and inflates every layout's absolute FoM by a factor of about 2.4–3.2; the project README quantifies the mismatch, and the ratios between the layouts are

unchanged by it. Until the covariance is rebuilt with the matching bias, the absolute values here are upper bounds of a rigid linear-bias model, and the content of this section is the comparison between bias priors on a fixed layout, which holds the layout, the data vector and the covariance fixed and changes only the prior.

Figure 6 places every scanned (L, σ_b) between its layout's two caps. At the reference point ($\sigma_b = 3\%$, $L = 0.05$) the four layouts give FoM = 11.8, 8.05, 2.01 and 0.779, which is 0.36, 0.40, 0.24 and 0.41 of the respective tied values. At $\sigma_b = 1\%$ the tied fraction spans 0.45–0.79 across layouts and correlation lengths. At $\sigma_b \geq 5\%$ the $\Delta d_{\rm com} = 62$ Mpc and $\Delta z = 0.05$ layouts sit near their independent values: most of the information the tying provided is already spent. Two scanned points exceed the LSST Y1 3x2pt reference of 17.3 (the dashed line), both on the $\Delta d_{\rm com} = 62$ Mpc layout at $\sigma_b = 1\%$: FoM = 18.7 at $L = 0.1$ and 20.8 at $L = 0.2$. The reference is the `lsst_y1` project's 3x2pt Fisher forecast in the DESC Science Requirements Document Year 1 setup [21], with the figure of merit defined as here.

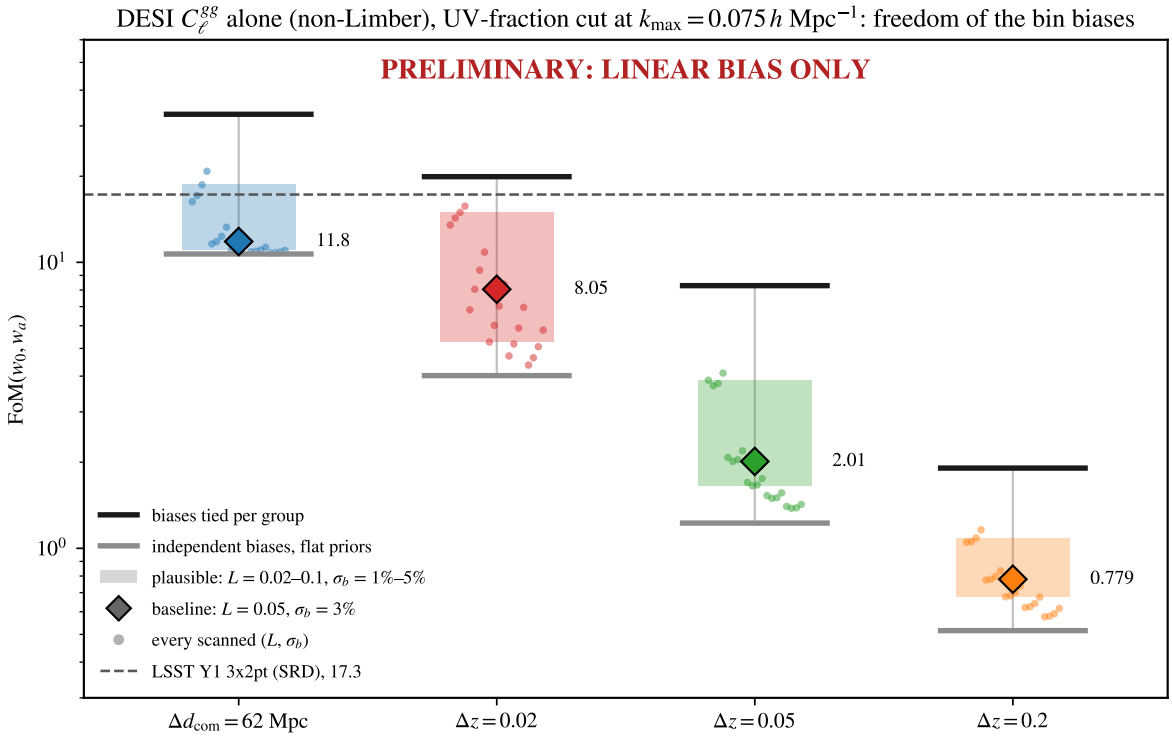


Figure 6: The FoM(w_0, w_a) of each thin-bin layout: the tied model (black cap), independent biases (gray cap), every scanned (L, σ_b) point (dots), the scanned range ($\sigma_b = 1$–5%, $L = 0.02$–0.1; the band the legend labels "plausible"), the kernel scan's reference point (the diamond the legend labels "baseline"), and the LSST Y1 3x2pt reference at 17.3 (dashed). The note's baseline is the memory model of Section 1, not a point of this scan.

Figure 7 resolves the same numbers on the (L, σ_b) grid. L matters where neighboring bins sit much closer than L, so that the correlation $e^{-\Delta z/L}$ stays high and lengthening L keeps converting bin-to-bin freedom back into effective tying. Both thin layouts are in that regime: the $\Delta d_{\rm com} = 62$ Mpc bins are $\Delta z = 0.016$–0.030 apart between $z = 0.2$ and 1.3, and on the $\Delta z = 0.02$ layout $e^{-\Delta z/L}$ runs from 0.37 at $L = 0.02$ to 0.90 at $L = 0.2$. At $\sigma_b = 1\%$ both

layouts respond: the $\Delta z = 0.02$ FoM rises from 13.5 to 15.7 across the L grid, and the $\Delta d_{\rm com}$ FoM by 28%, from 16.3 to 20.8. At $\sigma_b \geq 3\%$ the $\Delta z = 0.02$ layout keeps a strong response (6.8 to 10.9 at 3%) while $\Delta d_{\rm com} = 62$ Mpc sits close to its independent floor of 10.69, which no prior choice moves. In the two wide layouts the bins are as far apart as L or farther, the neighbor correlation is low everywhere on the grid, and L moves the FoM only weakly.

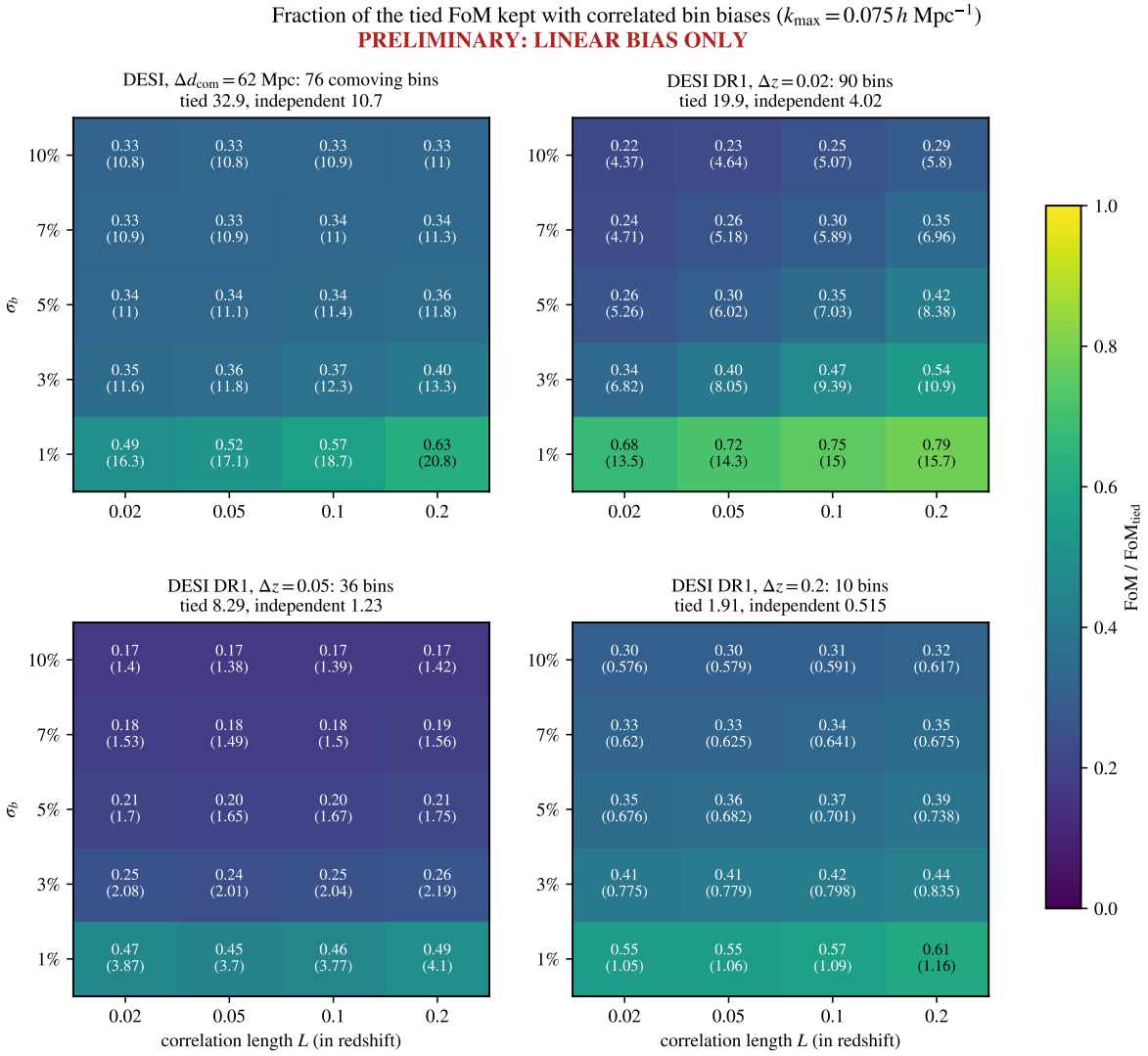


Figure 7: FoM/FoM$_{\rm tied}$ across the scanned (L, σ_b) grid, one panel per layout, with the absolute FoM in parentheses.

Figure 8 tests whether one number controls the cost. Its horizontal axis is $\sqrt{2}\,\kappa = \sigma_b\sqrt{2/L}$, the rms neighbor change per unit $\sqrt{\Delta z}$ of Eq. (12). The points do not collapse onto a single curve per layout. The reason is that the roughness controls only the short-separation jumps: at fixed κ, a longer L also carries a larger wander $\sigma_b = \kappa\sqrt{L}$, so the bias can drift coherently over the whole group, a freedom the roughness alone does not capture. The Fisher matrix pays for both kinds of freedom, so no single-variable description of the grid should be expected.

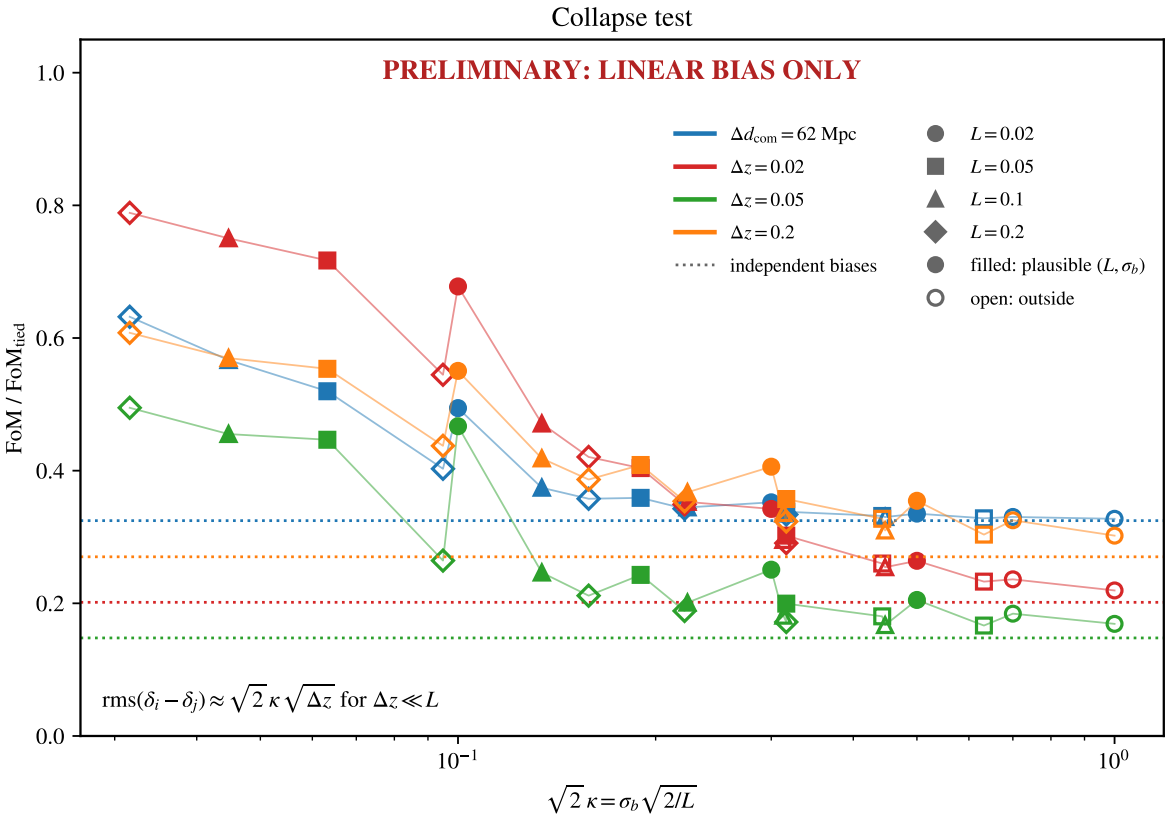


Figure 8: FoM/FoM$_{\rm tied}$ against the roughness combination $\sqrt{2}\,\kappa = \sigma_b\sqrt{2/L}$, which sets the rms neighbor change at small separations, $\sigma_b\sqrt{2\,\Delta z/L}$ (Eq. (12)): for $\sqrt{2}\,\kappa = 0.1$, the bias changes by 1% between bins $\Delta z = 0.01$ apart, and 1.4% for $\Delta z = 0.02$. Colors distinguish the four layouts, with each layout's independent fraction FoM$_{\rm indep}$/FoM$_{\rm tied}$ dotted; marker shapes give L (circle 0.02, square 0.05, triangle 0.1, diamond 0.2). Markers are filled when (L, σ_b) lies inside the scanned range, $\sigma_b = 1$–5% and $L = 0.02$–0.1 (the legend calls filled markers "plausible"), and open outside it. The range means inside the following edges, not a probability, and the stated assumptions of Section 1 supersede it as a prior choice. The lower edge $\sigma_b = 1\%$ is derived: a conserved population drifts from B/D by about 1% per $\Delta z = 0.05$ (Appendix C.2), so a narrower prior would be tighter than population evolution alone. The upper edge $\sigma_b = 5\%$ rests on measurements: the measured LRG residuals around B/D are at most 2.3%, with errors of 1–3.4%, at $\Delta z \approx 0.1$ spacing ([6]; Appendix C.5); the collected measurements of Appendix O place the upper envelope of the smooth LRG and ELG deviations at about 5%; and the neighbor-rms comparison of Appendix C.3 shows that $\sigma_b \geq 10\%$ lets neighboring bins differ by 8–27%, which no identified mechanism produces, so 10% and 20% are excluded from the region and 10% stays in the scan only as the edge of the grid. The upper edge $L = 0.1$ is a modeling choice: the argument of Appendix C.7 that a sample's flux limit and color cuts change the galaxy mix over $\Delta z \approx 0.1$, although the measured drifts are smooth and favor the long end of the L grid (Appendix C.6), the case the memory model of Appendix E addresses. The lower edge $L = 0.02$ is the thinnest bin spacing of the scanned layouts: below it, neighboring bins would decorrelate faster than the bins are spaced.

The scan reaches its two limits numerically, as Appendix J.4 says it must. At $\sigma_b = 0.01\%$ with $L = 0.05$ the four layouts give FoM = 32.93, 19.93, 8.285 and 1.907, against tied values 32.94, 19.93, 8.288 and 1.908; at $\sigma_b = 1000\%$ they land on the independent values 10.69, 4.016, 1.225 and 0.5154 exactly.

E The memory model

E.1 Why a second model

The kernel model carries one amplitude and one length, and Appendix C.6 found that the measured deviations are not the kind it describes best. Every deviation found, except the narrow sky-line windows, is a line-like drift across its group: the BGS flux limit drifts by 10–25% across the group, the LRG3 magnitude limit by 6–11% per $\Delta z = 0.1$, and a conserved population by about 1% per $\Delta z = 0.05$ (Appendix C.5).

The kernel model can admit such a drift, but only at a price. To pass a 20% drift with $L \leq 0.1$ it needs $\sigma_b = 10$–15%, and that spread also allows roughly 10% scatter between neighboring bins (Table 3), which no identified mechanism produces. One σ_b controls both the drift and the scatter, so the kernel cannot say "large smooth drift, small local scatter". The memory model separates the two. The left panel of Figure 9 shows what the kernel model assumes a galaxy population does; Figure 10 shows what the measured physics does.

E.2 Why the kernel forgets

The kernel's limitation is built into its correlation, $e^{-|z_i - z_j|/L}$, which decays with separation: at $L = 0.05$, bins 0.15 apart share $e^{-3} \approx 0.05$ of their deviation. Prediction makes the decay concrete. Appendix K shows that the best guess of the next bin given all earlier ones is

$$\mathrm{E}\big[\delta_{k+1} \mid \delta_1, \ldots, \delta_k\big] \;=\; r_k\, \delta_k : \tag{18}$$

only the nearest neighbor matters, and the influence of any one bin on the rest of the chain fades as the product of the neighbor correlations, roughly r^n after n steps. The kernel forgets.

E.3 The model, and why it remembers

The memory model adds one number per group: a tilt. The bias keeps the form of Eq. (4), and the deviation splits into a line and a residual,

$$b_i \;=\; \frac{B_g}{D(z_i)}\,\big(1 + \delta_i\big), \qquad \delta_i \;=\; s_g\, v_i + \varepsilon_i, \qquad v_i \;=\; z_i - \bar{z}_g, \tag{19}$$

where $\bar{z}_g$ is the mean of the group's bin redshifts, s_g is a free tilt per group with a flat prior (the same status as B_g), and the residual ε_i carries the exponential kernel of Eq. (5), $\langle \varepsilon_i \varepsilon_j \rangle = \sigma_b^2\, e^{-|z_i - z_j|/L}$.

To see what the tilt changes, give the slope a Gaussian prior of variance τ^2 first (the flat prior is the limit $\tau \to \infty$). The deviations then have covariance

$$\mathrm{Cov}(\delta_i, \delta_j) \;=\; \tau^2\, v_i\, v_j \;+\; \sigma_b^2\, e^{-|z_i - z_j|/L}. \tag{20}$$

The new term does not decay with separation. Bins on the same side of the group center ($v_i v_j > 0$) are positively correlated, bins on opposite sides negatively. For separations much larger than L the kernel term is gone and the correlation coefficient tends to

$$\rho_{ij} \;\longrightarrow\; \frac{\tau^2 v_i v_j}{\sqrt{\left(\tau^2 v_i^2 + \sigma_b^2\right)\left(\tau^2 v_j^2 + \sigma_b^2\right)}} \;\longrightarrow\; \pm 1 \quad \text{when } \tau|v| \gg \sigma_b : \tag{21}$$

the two ends of a group remember each other through the shared slope (Appendix L derives both steps).

Prediction shows the same memory. The best guess of the next bin becomes

$$\mathrm{E}\big[\delta_{k+1} \mid \delta_1, \ldots, \delta_k\big] \;=\; \hat{a} + \hat{s}_g\, v_{k+1} \;+\; r_k\,\big(\delta_k - \hat{a} - \hat{s}_g\, v_k\big), \tag{22}$$

where $(\hat{a}, \hat{s}_g)$ is the best-fitting straight line through all the past bins, weighted by C^{-1} (it needs at least two past bins). Every past bin now enters, through the fitted line; only the wiggle around the line stays Markov. Statistics calls Eq. (22) the universal-kriging predictor, prediction with an unknown linear mean; Appendix L proves it.

Figure 9 makes the difference visible on the BGS bins at $\Delta z = 0.02$. In the left panel, kernel draws at $\sigma_b = 10\%$, about what the kernel needs to admit a drift of BGS size, zigzag and forget: bins 0.1 apart are unrelated, and no draw looks like the thick flux-limit drift. In the right panel, memory draws with a tilt plus $\sigma_b = 2\%$ are lines with small wiggles, and the drift looks like a typical draw.

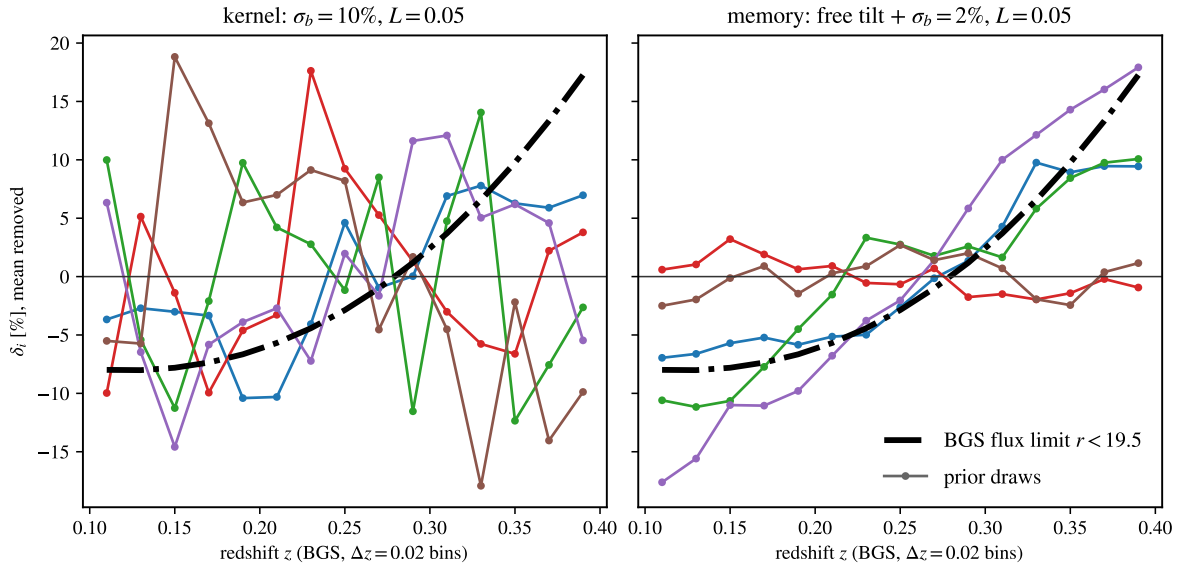


Figure 9: Random prior draws of the deviations of the 15 BGS bins at $\Delta z = 0.02$, each draw's mean removed (B_g is free). The thick dash-dot line is the BGS flux-limit drift. Left: the kernel model with $\sigma_b = 10\%$ and $L = 0.05$. Right: the memory model, a tilt plus $\sigma_b = 2\%$ and $L = 0.05$. The slope is a free parameter of the model; for display only, the drawn slopes come from a Gaussian of width 0.6 per unit redshift.

E.4 Why the memory model differs physically

A selected sample changes along a group in two physically different ways. The first kind is coherent: an evolution of the whole sample that moves every bin the same way. A flux or magnitude threshold that rises with distance, the debiasing of a conserved population, and the drift of the halo-mass threshold at fixed number density all act like this. The second kind is local: features of the selection that change bins independently and are forgotten over a short range, such as the sky-line windows and filter-edge crossings of Appendix C.4. The exponential kernel knows only the second kind, so it must inflate σ_b to admit the first. The memory model gives the first kind to the tilt and the second to the kernel.

One straight line is enough for the coherent kind because the drifts are smooth functions of redshift, set by $D(z)$ and the distance modulus, which change on Δz of order one. Over a group span of 0.2–0.5 the first-order Taylor term dominates. Table 4 measures this (derived): the best-fitting straight line through each group's conserved-population deviation, evaluated at the $\Delta z = 0.05$ bin centers with equal weights, and the residual the line leaves. A line removes the conserved-population drift to better than 0.3%.

Group	z range	span of δ	slope per unit z	residual rms	residual max
BGS	0.1–0.4	8.49%	−34.0%	0.062%	0.084%
LRG1	0.4–0.6	3.41%	−22.7%	0.020%	0.020%
LRG2	0.6–0.8	2.94%	−19.6%	0.018%	0.019%
LRG3	0.8–1.1	4.07%	−16.3%	0.038%	0.052%
ELG1	0.8–1.1	8.23%	−32.9%	0.077%	0.104%
ELG2	1.1–1.6	11.09%	−24.6%	0.158%	0.270%

Table 4: The best-fitting straight line through each group's conserved-population deviation $\delta(z)$ of Eq. (16), at the $\Delta z = 0.05$ bin centers with equal weights (derived; flat ΛCDM, $\Omega_m = 0.3$, amplitudes $B = 1.34$ for BGS, 1.70 for the LRGs and 0.84 for the ELGs, the reference at the group midpoint). The span is δ at the first center minus δ at the last; the residual columns describe what the line leaves.

The flux-limit drift is larger and not exactly straight. For BGS ($r < 19.5$, with r the apparent-magnitude band rather than a correlation coefficient; no K-correction; the threshold-bias fit of [12] with $M_* = -20.44$ used without evolution; magnitudes in the $M - 5\log_{10} h$ convention that matches M_*; and the group amplitude set so the mean deviation vanishes), the limiting magnitude runs from −18.56 to −21.24 between the first and last bin centers, the bias from 1.09 to 1.54, and the deviation spans 22.3% with a fitted slope of 89% per unit redshift. The line leaves 2.4% rms and 3.3% at most: the drift is convex, because the bias grows exponentially with the luminosity threshold (through the magnitude). So for BGS the memory model's residual scatter must be about 2–3%, or the model needs a quadratic term. Appendix N derives the flux-limit model step by step and lists what it leaves out: the threshold-bias fit frozen at $z \approx 0.1$, the missing K-correction and luminosity evolution, and the unmodeled fiber cut and BGS Faint sample.

Figure 10 carries both statements. In the top panel every conserved deviation is a straight line, and the thin gray fitted lines sit on top of them. In the bottom panel the conserved residuals are flat at zero, while the BGS flux limit leaves a ±3% U shape: that U is the curvature the memory model's σ_b must absorb.

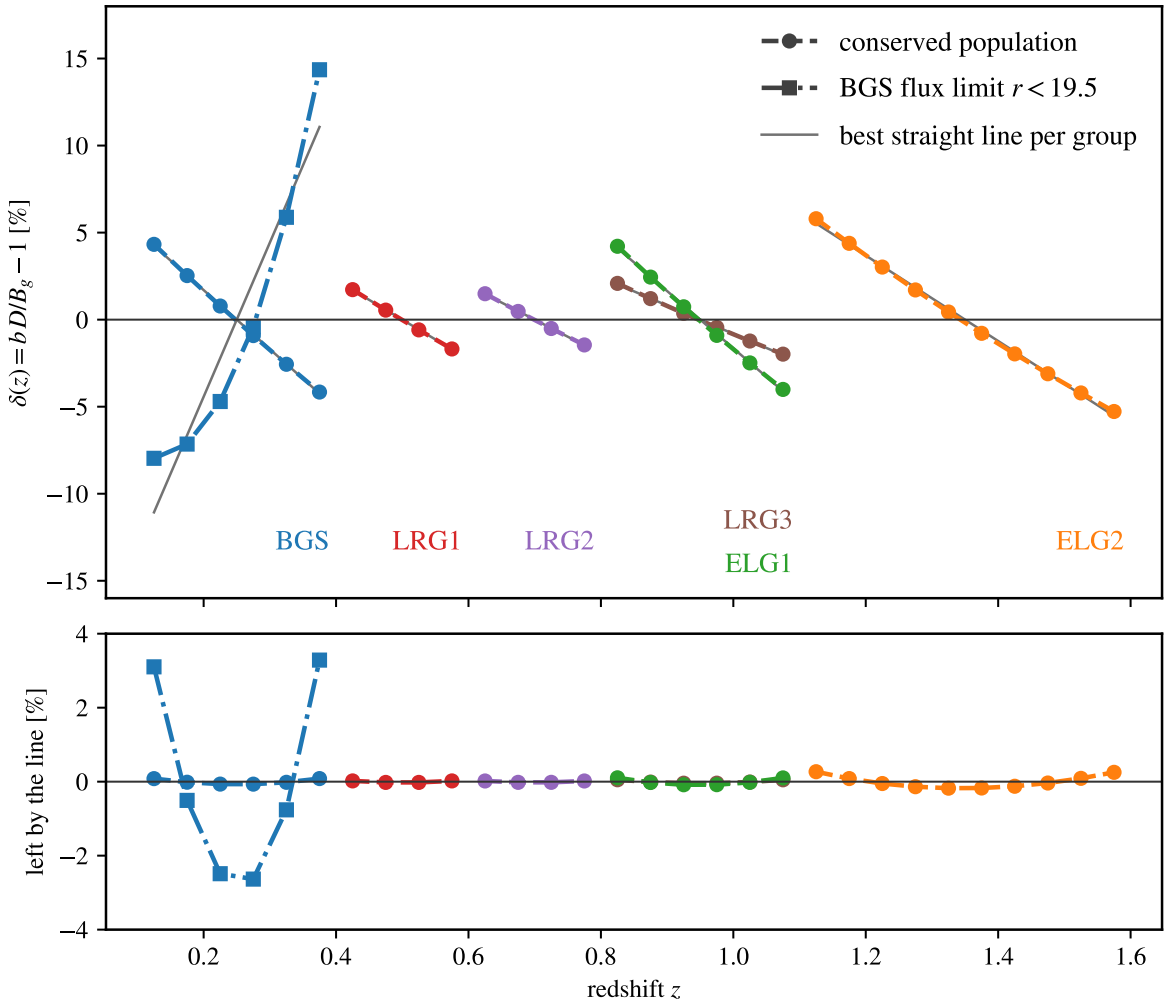


Figure 10: Top: the deviation $\delta(z) = b\,D/B_g - 1$ at the $\Delta z = 0.05$ bin centers, with B_g set so the mean deviation vanishes. Dashed with circles: the conserved population, every group. Dash-dot with squares: the BGS flux limit. Thin gray: the best straight line per curve. Bottom: what the line leaves. The conserved residuals are invisible (under 0.3%); the BGS flux-limit residual is convex, about ±3%. This is the figure behind Table 4.

The two models use the same symbol σ_b for different physical quantities. In the kernel model, σ_b is the total departure of a bin from the B_g/D trend: drift plus scatter. In the memory model, σ_b is the departure from the group's own straight line: curvature plus local scatter. The physically plausible values differ accordingly: about 0.1% for a conserved population (Table 4), 1–2% for the local selection features (Appendix C.5), and 2–3% for the BGS curvature.

The honest limit is unchanged: none of the measurements collected in Appendix C.5 resolves bias below $\Delta z \approx 0.1$. The memory model rests on the same evidence as the kernel model; it encodes that evidence in a form closer to the physics.

E.5 What the prior charges

With the flat priors restored (slope variance $\tau^2 \to \infty$, and amplitude variance $W^2 \to \infty$ for B_g), the prior precision of a group's bin amplitudes becomes

$$\mathsf{P}_{\rm mem} \;=\; \mathsf{Q} - \mathsf{Q}\,\mathsf{U}\big(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)^{-1}\mathsf{U}^{\mathsf{T}}\mathsf{Q}, \qquad \mathsf{Q} = \mathsf{C}^{-1}, \qquad \mathsf{U} = [\,\mathbf{1},\; v\,], \tag{23}$$

the same construction as Eq. (14) with one more column: U is the $n \times 2$ matrix whose columns are the ones vector and the centered redshifts v. The penalty it charges is

$$\delta^{\mathsf{T}}\mathsf{P}_{\rm mem}\,\delta \;=\; \min_{a,\,s_g} \big(\delta - a\,\mathbf{1} - s_g\,v\big)^{\mathsf{T}}\mathsf{Q}\big(\delta - a\,\mathbf{1} - s_g\,v\big)\,: \tag{24}$$

the prior subtracts the best-fitting straight line, weighted by C^{-1}, and charges only the residual. A monotonic drift of any size costs nothing; bumps around the line still cost, and still forget over the length L.

The structure of $\mathsf{P}_{\rm mem}$ separates local from global. Q is tridiagonal (Appendix J): a local object that links neighbors. The subtracted correction has rank two and is dense: it couples every bin to every other, because the fitted line listens to all of them. The kernel model already had one dense rank-one piece, the subtracted mean of Eq. (14); the memory model adds the slope. $\mathsf{P}_{\rm mem}$ annihilates both columns, $\mathsf{P}_{\rm mem}\mathsf{U} = 0$, is positive semidefinite, and has rank $n-2$ (Appendix L).

A group of one or two bins has $\mathsf{P}_{\rm mem} = 0$: an amplitude and a slope fix every bin, so the group is fully free. Among the scanned layouts this mostly affects $\Delta z = 0.2$, whose groups hold the fewest bins.

E.6 What the slope costs

The tilt has a physical price. In linear theory the clustering band powers of bin i scale as $b_i^2\,D(z_i)^2$, so the data measure the product $b\,D$, not the factors. Why a free tilt costs growth information takes one line: with $f = d\ln D/d\ln a$ the linear growth rate, $d\ln D/dz = -f(z)/(1+z)$, so inside a group

$$\frac{d\ln(bD)}{dz} \;=\; \frac{d\ln b}{dz} \;-\; \frac{f(z)}{1+z}. \tag{25}$$

A free tilt in b removes the channel that measures f through the amplitude evolution inside a group. The free amplitude B_g already handed each group's overall growth to the bias; the free tilt also hands over the relative growth inside a group.

Not everything is lost to the degeneracy. The redshift-space terms scale as $b\,f$, not as b^2, and measure f directly; the geometric distances and the shape of the power spectrum are untouched; and the comparison of growth between groups survives, because each group gives away only its own interior tilt, a statement this note does not test. This is why "tied plus free tilt" sits below the tied model, and why it should sit well above the independent one. The limit $\sigma_b \to 0$ of the memory model isolates the question: it measures how much of the figure of merit came from growth measured inside groups.

E.7 Results of the memory model

The memory model was scanned on the same grid as the kernel model: the same four layouts and the same FoM(w_0, w_a) of the DESI clustering band powers alone at $k_{\max} = 0.075\,h/\mathrm{Mpc}$, with σ_b and L now describing the residuals around each group's free line. The caveat of Appendix D applies unchanged: these results are preliminary, linear bias only.

Layout	tied	tied + free tilt	memory, 1%	kernel, 1%	independent
$\Delta d_{\mathrm{com}} = 62$ Mpc	32.94	24.35 (0.74)	14.99	17.12	10.69
$\Delta z = 0.02$	19.93	13.34 (0.67)	11.39	14.29	4.016
$\Delta z = 0.05$	8.288	2.654 (0.32)	1.683	3.701	1.225
$\Delta z = 0.2$	1.908	0.5154 (0.27)	0.5154	1.056	0.5154

Table 5: FoM(w_0, w_a) at $L = 0.05$. The tied-plus-free-tilt column is the $\sigma_b \to 0$ limit of the memory model, with its fraction of the tied value in parentheses; the memory and kernel columns are both at $\sigma_b = 1\%$.

The price of the tilt alone is the distance between the first two columns of Table 5. One free tilt per group costs 26% of the tied FoM on $\Delta d_{\mathrm{com}} = 62$ Mpc, 33% on $\Delta z = 0.02$, 68% on $\Delta z = 0.05$, and 73% on $\Delta z = 0.2$, where the free amplitude and tilt already free every one- and two-bin group completely. That is the information the tied model drew from the evolution of $b\,D$ inside each group, the growth channel of Eq. (25). $\Delta z = 0.05$ leans on that channel far more than the thinner layouts; a plausible reading, offered as interpretation rather than an established result, is that the thinner layouts measure more through the redshift-space terms and the cross-bin structure, which a tilt does not absorb.

At the same σ_b the memory model sits below the kernel model, as it must: it has strictly more freedom. In Figure 11 the dashed memory curves run below the solid kernel curves at small σ_b. That is not the comparison that matters.

The comparison that matters pairs each model with the σ_b it needs to describe the physics of Appendices C and E.4: about 10% for the kernel model to admit the measured drifts, and about 1% (a conserved, LRG-like population) to 3% (the BGS curvature) for the memory model. At $L = 0.05$, $\Delta d_{\mathrm{com}} = 62$ Mpc gives memory 14.99 at 1% and 11.45 at 3% against kernel 10.81 at 10%; $\Delta z = 0.02$ gives 11.39 and 7.41 against 4.64; and $\Delta z = 0.05$ gives 1.68 and 1.33 against 1.38. Once the measured drifts are admitted, the memory model keeps more of the FoM on the two thinnest layouts, and about the same on $\Delta z = 0.05$.

Figure 12 resolves the memory grid. As in the kernel model, L matters where neighbors sit much closer than L: at $\sigma_b = 3\%$ the $\Delta z = 0.02$ FoM runs from 6.1 at $L = 0.02$ to 10.0 at $L = 0.2$, and the $\Delta d_{\mathrm{com}} = 62$ Mpc layout responds at $\sigma_b = 1\%$ (14.0 to 18.2) before its independent floor takes over. $\Delta z = 0.2$ is degenerate by construction: every cell equals the

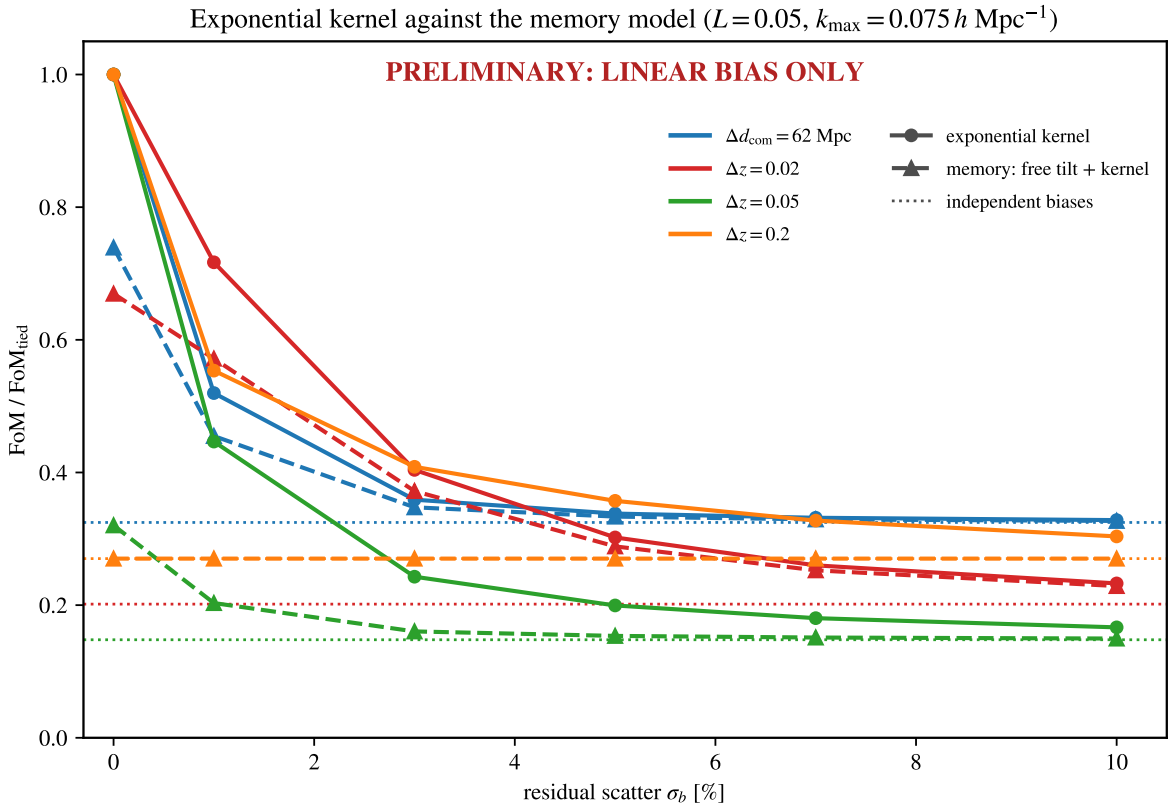


Figure 11: The two models against σ_b at $L = 0.05$, as the fraction of each layout's tied FoM. Solid lines with circles: the kernel model. Dashed lines with triangles: the memory model. Dotted: each layout's independent value. At $\sigma_b = 0$ the kernel model sits at one (the tied model) and the memory model at its tied-plus-free-tilt value. The dashed orange line of $\Delta z = 0.2$ lies on its dotted independent line.

independent value 0.5154, because each of its groups holds one or two bins, which the free amplitude and tilt free completely (Appendix E.5).

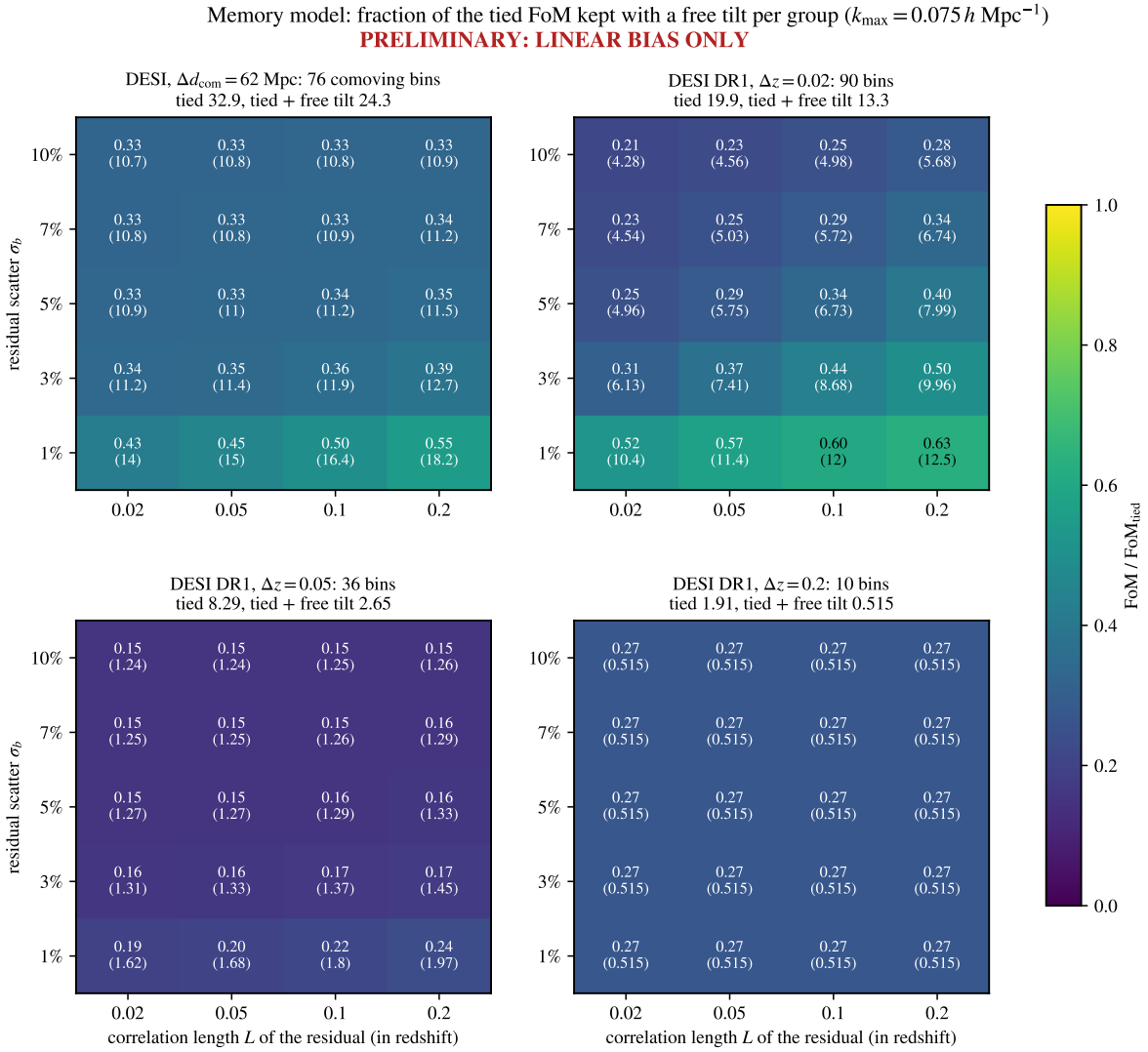


Figure 12: The memory model across the scanned (L, σ_b) grid, in the format of Figure 7: one panel per layout, with the absolute FoM in parentheses and each panel's tied-plus-free-tilt value in its title.

The scan reaches its limits numerically. At $\sigma_b = 1000\%$ the memory model lands on the independent values exactly (10.69, 4.016, 1.225, 0.5154), and at $\sigma_b \to 0$ on the tied-plus-free-tilt values of Table 5, as Appendix E.8 says it must.

E.8 The limits

The memory model contains the same two endpoints as the kernel model, shifted by the tilt. As $\sigma_b \to 0$ the residuals vanish and every group is a rigid line with a free level and a free slope: the tied model plus a free tilt per group. As $\sigma_b \to \infty$ the prior charges nothing and the biases

are independent. Between the endpoints, σ_b and L keep the same meanings as before, now applied to the wiggles around each group's line.

E.9 The two models in one paragraph

The intermediate bias model frees one amplitude B_g per tracer group and lets each bin deviate fractionally from the group's constant-amplitude trend $B_g/D(z)$, with deviations correlated over a redshift length L and a per-bin wander σ_b. In the Fisher matrix this becomes one projected precision P per group, added to the bin-amplitude block: P charges for differences between neighboring amplitudes but never for a uniform shift of the group. The unprojected inverse covariance links only neighboring bins directly (it is tridiagonal, even though the correlation itself reaches far); the projection adds one dense rank-one piece, the subtracted mean. The prior reduces to the tied model as $L \to \infty$ and to independent flat-prior biases as $\sigma_b \to \infty$. A second model adds a free tilt per group and reserves σ_b for the residuals around each group's line: the tilt alone costs 26–73% of the tied FoM, and at the σ_b each model needs to admit the measured drifts (about 1–3% with a tilt, about 10% without), the memory model keeps more of the figure of merit on the thinnest layouts.

F The accuracy test: the figure of bias

The ladder of Section 1 orders the models by how much information they keep. This appendix orders them by the other quantity that matters: whether each model's quoted error still covers the truth when the true biases carry structure the model cannot express. It injects the two known drifts first, then random truths built on them, and ends at the error inflation, the factor by which each rung's (w_0, w_a) ellipse is too small. The result is the accuracy statement the summary quotes, and the reason the tied model is excluded rather than merely optimistic.

F.1 The shift formula

Let m be the expected data vector (the band powers), C_d its covariance, D the matrix of derivatives of m with respect to the parameters, and $F = \mathsf{D}^\mathsf{T}\mathsf{C}_d^{-1}\mathsf{D} + F_{\text{priors}}$ the Fisher matrix of one rung, priors included (Appendix P). Now let the true biases carry a known drift the rung's model may not be able to express: the data vector moves by δm. To first order in δm, the maximum of the posterior moves by

$$\delta p \;=\; F^{-1}\mathsf{D}^\mathsf{T}\mathsf{C}_d^{-1}\,\delta m, \tag{26}$$

the linear response of the fit. The derivation is one line: the score (the gradient of the log posterior) at the old maximum becomes $\mathsf{D}^\mathsf{T}\mathsf{C}_d^{-1}\delta m$ instead of zero, and the curvature that converts a score into a parameter displacement is F.

The quoted number compresses δp into the plane that matters. With C_w the rung's marginalized 2×2 covariance of (w_0, w_a) and δw the (w_0, w_a) entries of δp, the shift is

$$\Delta_w \;=\; \sqrt{\delta w^\mathsf{T}\,\mathsf{C}_w^{-1}\,\delta w}\,: \tag{27}$$

the displacement of the dark-energy ellipse in units of its own size. $\Delta_w = 1$ means the fitted (w_0, w_a) sits on the one-sigma contour of the unshifted forecast. The threshold drawn in the

figures is $\Delta_w = 1/3$, a shift that widens the effective error by a few percent. Both the ellipse and the response use the same rung's F: a rigid model is punished twice, because it absorbs little of the drift and its ellipse is small.

The drift enters through the bin biases. With β_i the bin amplitudes of Eq. (6) and δ_i the drift's fractional deviations, the data move by $\delta m = \sum_i (\partial m/\partial \beta_i)\, B_g\, \delta_i$, with each group's mean deviation removed because B_g is free. For the tied rung the response is evaluated in the tied parameter basis; for the independent rung the bias derivatives span the drift exactly, so the cosmology shift vanishes: the free biases absorb the drift, and the accuracy of that rung is bought entirely with the information the ladder shows it giving up.

F.2 The drift laws

Two known drifts are injected, the two derived in this note. The conserved-population drift applies to every group: the deviation of Eq. (16), anchored at the group's mean redshift. It is the gentlest drift physics supplies, nearly a straight line of a few percent across each group. The BGS flux-limit drift applies to BGS alone: the convex deviation of Appendix N, a 22% span whose best line leaves a 2.4% rms curvature. Both come from the same scripts as the theory figures; only the shape of each drift matters, because the free amplitudes remove the level.

F.3 The drift-only shifts

Table 6 lists the shift Δ_w of every rung when the data carry exactly one known drift and nothing else.

Drift	Layout	tied	baseline	pessimistic	independent
conserved population	$\Delta d_{\rm com} = 62$ Mpc	17.4	0.027	0.003	0
(every group)	$\Delta z = 0.02$	12.4	0.038	0.005	0
	$\Delta z = 0.05$	21.5	0.097	0.008	0
	$\Delta z = 0.2$	10.9	0	0	0
BGS flux limit	$\Delta d_{\rm com} = 62$ Mpc	0.13	0.0014	0.0013	0
($r < 19.5$)	$\Delta z = 0.02$	1.15	0.022	0.024	0
	$\Delta z = 0.05$	2.79	0.27	0.29	0
	$\Delta z = 0.2$	1.39	0	0	0

Table 6: The (w_0, w_a) shift Δ_w of Eq. (27), in units of each rung's own marginalized ellipse, when the data carry one of the two known drifts. On $\Delta z = 0.2$ the baseline and pessimistic rungs equal the independent model (every group has one or two bins), so their shifts are zero.

Two statements carry the table. First, the tied model is excluded on accuracy. The conserved population, the gentlest drift, biases it by 10.9 to 21.5 ellipse units; its flux-limit column is smaller only because that drift moves one group out of six. A model whose central value sits more than ten of its own sigmas from the truth under the mildest input is not a conservative choice, it is a wrong one, and this is independent of the ladder's information argument. Second, the baseline and the pessimistic rung absorb both known drifts. The largest entry of either is the BGS flux limit on $\Delta z = 0.05$, about 0.3: that layout resolves BGS into only six bins, so the convex residual is sampled coarsely and the prior has less room to absorb it between the free line and the data.

A known drift is only half of a bias history. The two drifts are the deviations this note can derive; a real sample adds residual structure whose shape nobody has measured. The rest of this appendix makes that structure random and asks each rung the sharper question: does the error ellipse you quote still cover the truth?

F.4 Random truths

A random truth sets the bias of bin i in group g to

$$b_i \;=\; \frac{B_g}{D(z_i)}\left(1+\delta_i^{\rm drift}+\varepsilon_i\right) \tag{28}$$

at the fiducial cosmology. The drift $\delta_i^{\rm drift}$ is the known part: the $r<19.5$ flux-limit deviation in BGS and the conserved-population deviation of Eq. (16) in the other five groups. The residual ε_i is the random part: one draw from the Gaussian $\mathcal{N}(0,\Sigma)$ whose covariance is the kernel of one of the two stated priors, $\Sigma_{ij}=\sigma_g^2\,e^{-|z_i-z_j|/L}$ inside a group and zero across groups. Two families of 2000 truths are drawn: baseline scatter ($\sigma_g=3\%$ for BGS, 1% for the other groups, $L=0.2$) and pessimistic scatter ($\sigma_g=3\%$ in every group, $L=0.05$).

Every truth is fitted with every rung, and every rung's bias prior stays centered on the tied shape $\delta=0$. No rung is told which truth it faces: the baseline rung meets truths drawn from its own prior (the matched case below) and truths drawn from the pessimistic prior, and is judged on both.

The sampling costs one matrix product because the response is linear. Eq. (26) is linear in δm, and δm is linear in the bias deviations, so the (w_0,w_a) rows of the response collapse into one $2\times n_{\rm bins}$ matrix A per rung and layout:

$$\delta w \;=\; \mathsf{A}\,\delta, \qquad \delta_i \;=\; \delta_i^{\rm drift}+\varepsilon_i. \tag{29}$$

The draws also check themselves against the closed form they must reproduce, $\mathrm{E}[\Delta_w^2]=\Delta_{\rm drift}^2+\mathrm{tr}(\mathsf{C}_w^{-1}\mathsf{A}\Sigma\mathsf{A}^{\mathsf{T}})$, where $\Delta_{\rm drift}$ is the shift of the drift alone (Table 6) and C_w is the rung's marginalized (w_0,w_a) covariance.

One caveat bounds the method. The response is first order in δ, and the band powers are quadratic in the bin biases, so a term of order δ^2 is dropped: a few percent of the shift for the 1–3% residuals, and perhaps 10–20% of it for the 22% BGS drift. No conclusion of this appendix depends on that term.

Figure 2, in the summary (Section 1), draws the result; its caption is written to stand alone. The one subtlety is the matched rung, the gray column labeled "fit prior = truth scatter". Where the fit's prior equals the truth's scatter, the average shift is not a bias. That rung's ellipse is wider than the tied model's precisely because it marginalizes over the scatter the truth now actually has: the fitted (w_0,w_a) moves from draw to draw, and the ellipse was built to cover those moves. A correctly specified model therefore shows a nonzero average shift just by being correct, and the mean of Δ_w on the matched rung measures the share of the ellipse the bias scatter occupies, not an error of the model. Deciding whether the quoted ellipse is honest needs the statistic of the next subsection.

F.5 The error inflation

A rung is correctly specified when its quoted ellipse covers the truth with the stated probability. The statistic that tests exactly that is the squared distance χ^2_w of the fitted (w_0,w_a) from the

truth in units of the rung's ellipse, with every source of displacement counted: the drift, the bias scatter, and the data noise. Over the truths and the noise its expectation is

$$\mathrm{E}[\chi^2_w] \;=\; \Delta^2_{\mathrm{drift}} \;+\; \mathrm{tr}\Big(\mathsf{C}_w^{-1}\mathsf{A}\,\Sigma\,\mathsf{A}^{\mathsf{T}}\Big) \;+\; \mathrm{tr}\big(\mathsf{C}_w^{-1}\mathsf{N}_w\big)\,, \tag{30}$$

where N_w is the share of C_w contributed by the data noise and by the Gaussian priors on the per-group velocity dispersions σ_v, the only priors besides the bias prior. The quoted number is the error inflation

$$\sqrt{\mathrm{E}[\chi^2_w]\,/\,2}. \tag{31}$$

For each fitted rung, $\mathrm{E}[\chi^2_w]$ is the expected squared distance between the fitted (w_0, w_a) and the truth, in units of the ellipse C_w that the rung reports. Eq. (30) splits it into three sources, in order: the drift squared, the scatter the prior did not absorb, and the noise.

The drift term $\Delta^2_{\mathrm{drift}}$ is the squared shift from the known drift alone, the same for every truth. The scatter term carries the truth's residual ε into (w_0, w_a) through the response matrix A of Eq. (29): it is the part of the scatter that the rung's bias parameters, held back by their prior, did not follow. The noise term counts the data noise and the σ_v priors.

An honest ellipse gives exactly 2, the number of parameters. Suppose the rung is correctly specified: no drift remains, and the fitted (w_0, w_a) scatters around the truth with covariance exactly C_w. The expected squared distance is then $\mathrm{tr}(\mathsf{C}_w^{-1}\mathsf{C}_w)$, the trace of the 2×2 identity matrix, so $\mathrm{E}[\chi^2_w] = 2$: one unit for w_0 and one for w_a. A larger value means the fit lands, on average, farther from the truth than the reported ellipse allows.

The error inflation of Eq. (31) turns that excess into a length. Growing both axes of the ellipse by a factor f replaces C_w by $f^2\,\mathsf{C}_w$ and divides χ^2_w by f^2, so $f = \sqrt{\mathrm{E}[\chi^2_w]/2}$ is the growth that brings the expectation back to 2. The inflation is the factor by which the axes of the reported ellipse would have to grow to cover the truth as often as the ellipse claims: 1 for an honest ellipse, above 1 for an ellipse that is too small, below 1 for a conservative one.

The factor matches the average squared distance. It matches the coverage exactly when the real scatter of the fitted (w_0, w_a) has the shape and orientation of the reported ellipse.

Worked case: the baseline fit of a pessimistic truth. The baseline prior states that each group's residual ε around its free line (the free amplitude B_g and tilt s_g of Eq. (1)) is small and smooth. Its width, the rms σ_g of one bin's residual, is 1% (3% for BGS), and its correlation length $L = 0.2$ keeps the residual coherent over $\Delta z \approx 0.2$. The pessimistic truth has residuals of $\sigma_g = 3\%$ with $L = 0.05$ in every group: 3% wiggles that change over $\Delta z \approx 0.05$.

The baseline prior charges a large penalty for bin biases that follow such wiggles, so the fit lets the biases follow them only in part. The band powers still carry the rest of each wiggle, and the fit moves the cosmological parameters, (w_0, w_a) among them, to absorb what the biases do not.

The push on (w_0, w_a) is not one fixed offset. The drift part alone is $\Delta_{\mathrm{drift}} = 0.03$ on the $\Delta d_{\mathrm{com}} = 62$ Mpc layout. The rest comes from the residual and changes from truth to truth: over the 2000 truths the shift Δ_w of Eq. (27), in units of the baseline ellipse, lies between 0.7 and 4.0 for the central 68% (the 16th to 84th percentiles). The observed universe is one draw: one of these displacements, reported with the baseline ellipse.

Adding the noise term, Eq. (30) gives $\mathrm{E}[\chi^2_w] = 10.4$ on $\Delta d_{\mathrm{com}} = 62$ Mpc, an inflation of $\sqrt{10.4/2} = 2.28$ (Table 7). The real scatter of the best-fit (w_0, w_a) around the truth is about

2.3 times wider per axis than the reported ellipse. The contour the baseline reports as one sigma is the 0.44σ contour of the real scatter ($1/2.28 = 0.44$).

The figure of merit follows the same factor. $\mathrm{FoM}(w_0, w_a) = [\det \mathsf{C}_w]^{-1/2}$ is proportional to the inverse area of the ellipse; growing both axes by f multiplies the area by f^2, so the FoM scales as $1/f^2$. The baseline FoM of 18.19 on $\Delta d_{\rm com} = 62$ Mpc (Table 1), whose ellipse axes are 2.28 times too small, corresponds to $18.19/2.28^2 \approx 3.5$.

That value is not a number to quote: it rescales an ellipse whose center and shape still come from a prior narrower and smoother than the truth's scatter. If the pessimistic truth is possible, the fit to report is the pessimistic rung: its inflation on the same truth is 1.00, and its FoM is 11.45.

The opposite mismatch errs on the conservative side. A truth with baseline scatter, fitted with the pessimistic rung, gives an inflation of 0.87 to 0.99 on the $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$ and $\Delta z = 0.05$ layouts (Table 8). The real scatter is then narrower than the reported ellipse, which is at most 15% wider per axis than needed ($1/0.866 = 1.15$).

Layout	$\mathrm{E}[\chi^2_w]$	error inflation $\sqrt{\mathrm{E}[\chi^2_w]/2}$
$\Delta d_{\rm com} = 62$ Mpc	10.4	2.28
$\Delta z = 0.02$	4.66	1.53
$\Delta z = 0.05$	9.15	2.14
$\Delta z = 0.2$	2.00	1.00

Table 7: The worked case: the baseline rung fitted to truths that carry the known drifts plus residual scatter drawn from the pessimistic prior ($\sigma_g = 3\%$ in every group, $L = 0.05$). $\mathrm{E}[\chi^2_w]$ is the expected squared distance of Eq. (30) between the fitted (w_0, w_a) and the truth, in units of the baseline ellipse, with the data noise and the σ_v priors included; an honest ellipse gives 2. The error inflation of Eq. (31) is the factor by which the ellipse axes would have to grow to cover the truth as often as claimed. On $\Delta z = 0.2$ each group has one or two bins, so the free amplitude and tilt absorb every bias deviation and the inflation is 1.00.

The matched case lands on 1 by an identity, not by tuning. The matched rung's bias prior is the projected precision P of Appendix L, built from the same kernel covariance Σ the truth's scatter uses, and every projection of that form obeys $\mathsf{P}\,\Sigma\,\mathsf{P} = \mathsf{P}$. That identity makes the scatter term and the noise term of Eq. (30) sum to exactly 2: the widening the prior adds to the ellipse equals the error the scatter inflicts, and the rest of the ellipse is the noise the data carry anyway. The matched inflation is then $\sqrt{1 + \Delta^2_{\rm drift}/2} \approx 1$, because the matched rungs absorb the known drifts to $\Delta_{\rm drift} \lesssim 0.3$ (Table 6).

Three statements carry Table 8. First, the tied model is wrong under any truth: its ellipse is 8.5–18 times too small, the coverage form of the exclusion of Appendix F.3. Second, the baseline is correct exactly when nature's local bias features are no larger than its own prior allows: 1.00–1.01 under its own scatter, but 1.5–2.3 times too small under the pessimistic truth. The baseline FoM of Table 1 therefore rests on that assumption, and the pessimistic FoM is the safe statement. Third, the pessimistic rung is correct (1.00–1.02) or conservative (0.87–0.99) under both truths: a prior wider than the truth wastes information, which the ladder prices, but never understates the error. One qualification remains: scatter larger or rougher than the pessimistic prior would indict the pessimistic rung in the same way, and against that the ladder ends at the independent model, whose inflation is 1.00 for every truth expressible through the

Truth scatter	Layout	tied	baseline	pessimistic	independent
baseline	$\Delta d_{\rm com} = 62$ Mpc	12.5	1.00	0.971	1.00
	$\Delta z = 0.02$	8.54	1.00	0.866	1.00
	$\Delta z = 0.05$	16.9	1.01	0.994	1.00
	$\Delta z = 0.2$	8.82	1.00	1.00	1.00
pessimistic	$\Delta d_{\rm com} = 62$ Mpc	13.3	2.28	1.00	1.00
	$\Delta z = 0.02$	8.89	1.53	1.00	1.00
	$\Delta z = 0.05$	17.6	2.14	1.02	1.00
	$\Delta z = 0.2$	9.96	1.00	1.00	1.00

Table 8: The error inflation $\sqrt{\mathrm{E}[\chi^2_w]/2}$ of Eq. (31): the factor by which each rung's quoted (w_0, w_a) ellipse is too small when the truth carries the known drifts plus residual scatter drawn from the named prior. An entry of 1.00 means the ellipse covers the truth as it claims; an entry below 1 means the ellipse is conservative. On $\Delta z = 0.2$ every group holds one or two bins, so the three lower rungs absorb any truth and sit at 1.00.

bin biases.

F.6 The same test for Ω_m and A_s

The statistics above score the dark-energy pair (w_0, w_a). The same known drifts and the same 2000 random truths per case, drawn with the same random seed, are scored here for the matter density Ω_m and the primordial scalar amplitude A_s, each in units of its own marginalized error σ_p. For one parameter p the squared distance of the fit from the truth is $\chi^2_p = (\hat{p} - p_{\rm true})^2/\sigma^2_p$, with $\hat{p}$ the fitted value; an honest error bar gives $\mathrm{E}[\chi^2_p] = 1$, so the error inflation is $\sqrt{\mathrm{E}[\chi^2_p]}$: 1 for an honest error bar, above 1 the factor by which the bar is too small. For a matched case the identity $\mathsf{P}\,\Sigma\,\mathsf{P} = \mathsf{P}$ makes the scatter and noise terms sum to exactly 1, so the inflation is $\sqrt{1 + \Delta^2_{{\rm drift},p}}$ and departs from 1 only through the known drift. As a check that the truths are the same, the matched cases reproduce the (w_0, w_a) inflation of Table 8, within 0.019 of 1.

The known drifts alone, scored this way, repeat the pattern of Table 6. The conserved-population drift moves the tied rung by 3.99, 5.21, 11.7 and $6.42\,\sigma$ in Ω_m and by 3.79, 9.90, 5.82 and $0.58\,\sigma$ in A_s, in the layout order $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$, 0.05, 0.2. The baseline and pessimistic rungs stay at or below $0.29\,\sigma$; the largest entries are the BGS flux limit on $\Delta z = 0.05$, at 0.26–$0.28\,\sigma$ in Ω_m. Table 9 gives the error inflation with random truths.

The (w_0, w_a) verdict holds for Ω_m and A_s. The tied rung is excluded. The baseline is honest under its own scatter, but its Ω_m error bar is up to 2.75 times too small if local features reach the pessimistic size. The pessimistic rung is honest. On some layouts Ω_m and A_s react more strongly than (w_0, w_a) (the baseline rung under pessimistic scatter on $\Delta z = 0.05$: 2.75 in Ω_m against 2.14), because a bias error moves the clustering amplitude $b_i^2 A_s D^2(z_i)$, which A_s scales directly and Ω_m moves through the growth factor.

Truth scatter	Fit rung	Ω_m	A_s
baseline	tied	4.17 / 4.46 / 14.0 / 8.15	3.93 / 9.48 / 4.84 / 1.25
	baseline	1 / 1 / 1 / 1	1 / 1 / 1 / 1
	pessimistic	0.96 / 0.87 / 0.99 / 1.00	0.98 / 0.94 / 0.99 / 1.00
pessimistic	tied	5.87 / 5.18 / 14.5 / 8.96	4.61 / 9.76 / 5.39 / 2.93
	baseline	2.09 / 1.29 / 2.75 / 1.00	1.62 / 1.16 / 1.96 / 1.00
	pessimistic	1.00 / 1.00 / 1.04 / 1.00	1.00 / 1.00 / 1.02 / 1.00

Table 9: The error inflation $\sqrt{\mathrm{E}[\chi^2_p]}$ of Ω_m and A_s, each in units of its own marginalized error, when the truth carries the known drifts plus residual scatter drawn from the named prior. An entry of 1 means the error bar covers the truth as it claims. Each cell lists the layouts $\Delta d_{\mathrm{com}} = 62$ Mpc / $\Delta z = 0.02$ / $\Delta z = 0.05$ / $\Delta z = 0.2$. The truths are those of Table 8. The independent rung absorbs every such truth exactly and is omitted.

G The roughness scan

G.1 One number controls smooth deviations

For deviations that change slowly from bin to bin, the exponential kernel of Eq. (5) does not constrain σ_b and L separately. The jump variance of Eq. (11) depends on them only through σ_b^2/L, so a scan in two parameters hides a one-parameter family: doubling L while multiplying σ_b by $\sqrt{2}$ admits the same smooth drift. The combination this appendix uses is the roughness

$$x \;\equiv\; \sigma_b \sqrt{\frac{0.1}{L}} \;=\; \sigma_b \sqrt{\frac{2 \times 0.05}{L}}, \tag{32}$$

the small-separation rms change of the deviation between two bins $\Delta z = 0.05$ apart, a rescaling of the κ of Eq. (12) into an rms with a definite separation.

The roughness corrects a statement made in Appendix C. The argument there reads a short L as unphysical because no selection mechanism decorrelates neighboring bins that fast. That was too strong: $L = 0.05$ with the matching σ_b admits the real smooth drifts perfectly well, since only x matters for them. What a short L also admits is jagged histories nobody has observed, and what it cannot do is admit a large drift and forbid the jagged histories at the same time. The scan below makes that trade visible.

G.2 The scan

Figure 13 scans both models against x on a logarithmic axis, with one width σ_b for every group, at every correlation length L in $\{0.05,\ 0.1,\ 0.2,\ 0.5,\ 1\}$. The top row is the fraction of the tied FoM kept; the bottom row is the conserved-drift shift Δ_w of Appendix F. The left column is the exponential kernel alone (a free amplitude per group); the right column is the memory model (a free amplitude and tilt per group).

The memory curves collapse in L: at fixed x, every L gives nearly the same FoM fraction and the same drift shift, because the free tilt owns the whole-group freedom and the residual kernel acts only through its small-separation behavior, which is x. For the memory model the roughness is the model, and the stated assumptions are three marks on one axis: baseline $x = 0.71\%$ for the non-BGS groups, baseline $x = 2.1\%$ for BGS, pessimistic $x = 4.2\%$.

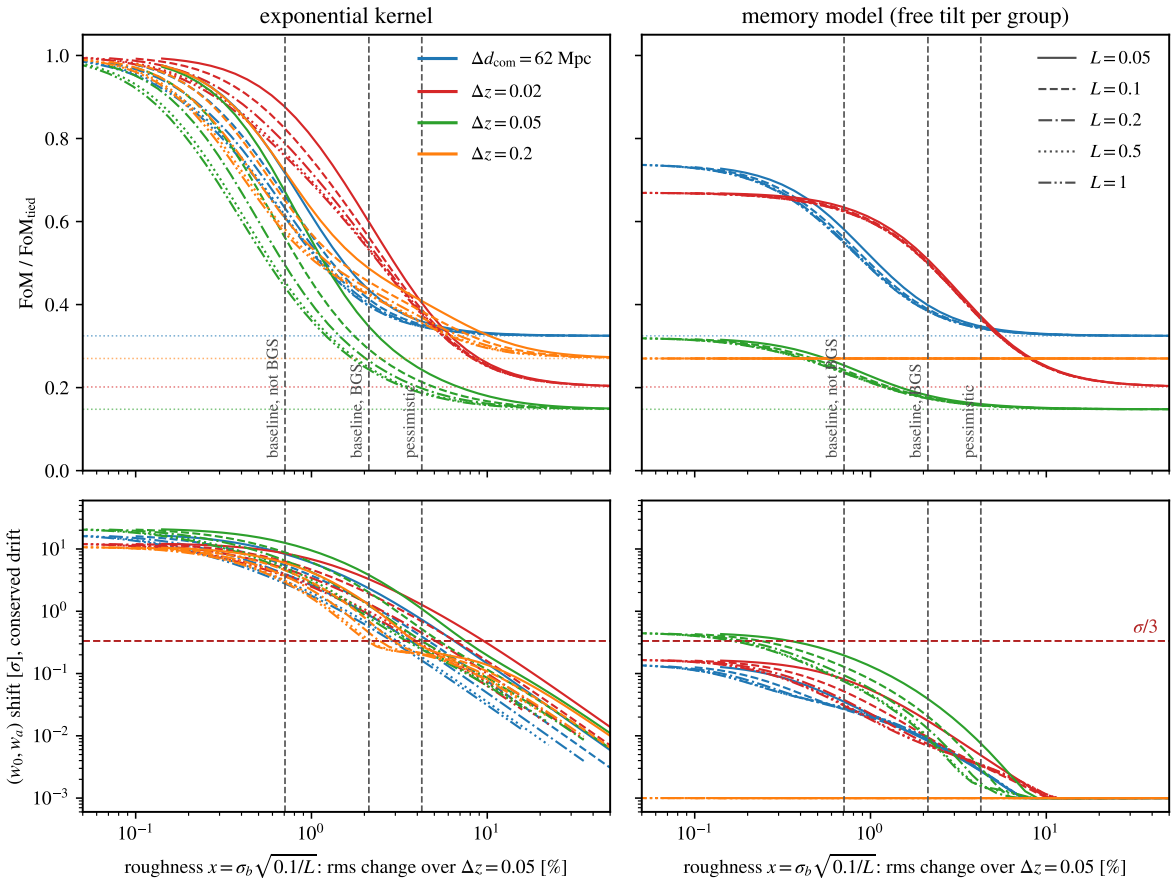


Figure 13: Both models against the roughness $x = \sigma_b\sqrt{0.1/L}$, the rms change of the bias deviation between bins $\Delta z = 0.05$ apart, with one σ_b for every group and every $L \in \{0.05, 0.1, 0.2, 0.5, 1\}$ (line styles). Left: the exponential kernel (free amplitude per group). Right: the memory model (free amplitude and tilt per group). Top: the fraction of the tied FoM kept, with each layout's independent fraction dotted. Bottom: the conserved-drift shift Δ_w in units of each model's own (w_0, w_a) ellipse, with the $\sigma/3$ threshold dashed; values below 10^{-3} are drawn at 10^{-3}. The vertical marks are the stated assumptions: baseline $x = 0.71\%$ (the non-BGS groups), baseline BGS $x = 2.1\%$, and pessimistic $x = 4.2\%$.

The kernel curves spread in L. At fixed x a longer L carries a larger wander $\sigma_b = x\sqrt{L/0.1}$, so the bias may drift coherently across the whole group, a freedom the roughness does not measure. The spread is the two-parameter nature of the kernel made visible, and it is why Appendix D finds no collapse onto a single curve.

The bottom row states the accuracy trade. The kernel needs a roughness of about 3–10%, depending on layout and L, before the conserved-drift shift falls under the threshold: it must buy accuracy with bin-to-bin scatter that no measurement supports. The memory model sits under the threshold everywhere except the tilt-only limit ($x \to 0$) on $\Delta z = 0.05$, which reaches 0.46: one straight line per group absorbs the conserved drift to a third of the ellipse or better once any residual freedom at all is granted.

H The projected precision for two bins

Everything essential already happens with two bins. Call their amplitudes β_1 and β_2, with common fiducial value B, prior spread s per bin, and correlation coefficient r between the two deviations. The prior covariance and its inverse are

$$\mathsf{C} \;=\; s^2 \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}, \qquad \mathsf{C}^{-1} \;=\; \frac{1}{s^2\,(1-r^2)} \begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix}. \tag{33}$$

(The inverse is checked by multiplying: the off-diagonal of $\mathsf{C}\,\mathsf{C}^{-1}$ is $r - r = 0$ and the diagonal is $(1-r^2)/(1-r^2) = 1$.)

Now examine the prior penalty $\chi^2_{\rm prior} = \Delta\beta^{\mathsf{T}}\,\mathsf{C}^{-1}\Delta\beta$, where $\Delta\beta = (\beta_1 - B,\ \beta_2 - B)^{\mathsf{T}}$ is the displacement from the fiducial. Two kinds of move matter.

A uniform shift. Take $\Delta\beta = a\,(1,1)^{\mathsf{T}}$: both bins move up by the same amount a. Working out the quadratic form with Eq. (33),

$$\Delta\beta^{\mathsf{T}}\mathsf{C}^{-1}\Delta\beta \;=\; \frac{a^2}{s^2(1-r^2)}\,\big[\,(1-r)+(1-r)\,\big] \;=\; \frac{2a^2}{s^2\,(1+r)} \;\neq\; 0. \tag{34}$$

The plain prior charges for a uniform shift. But a uniform shift of all β_i is a change of B_g, a direction the tied model leaves completely free. If the intermediate model kept this penalty, it would constrain B_g and the comparison between the models would be unfair.

A shape change. Take $\Delta\beta = a\,(1,-1)^{\mathsf{T}}$: the bins move apart. The same algebra gives

$$\Delta\beta^{\mathsf{T}}\mathsf{C}^{-1}\Delta\beta \;=\; \frac{2a^2}{s^2\,(1-r)}. \tag{35}$$

This is the penalty the prior is *meant* to apply: it grows without bound as $r \to 1$, where the two bins are forced to track each other, and it relaxes to the independent value $2a^2/s^2$ as $r \to 0$.

A number makes the contrast concrete. At $r = 0.670$ (the worked value of Appendix B.2), the shape change costs $2a^2/(0.330\,s^2) = 6.1\,a^2/s^2$ while the uniform shift costs $2a^2/(1.670\,s^2) = 1.2\,a^2/s^2$. The correlated prior already charges five times more for the shape change than for the uniform shift; the projection below takes the uniform-shift charge all the way to zero.

The projected precision. The repair for the uniform shift is the projection P of Eq. (14). Appendix I explains where the formula comes from; here, compute it for two bins. Because the two rows of C^{-1} have the same sum, $\mathsf{C}^{-1}\mathbf{1} = \big[s^2(1+r)\big]^{-1}(1,1)^{\mathsf{T}}$ and $\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1} = 2/\big[s^2(1+r)\big]$. Substituting into Eq. (14) and simplifying each entry,

$$\mathsf{P} \;=\; \frac{1}{2s^2(1-r)}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \Delta\beta^{\mathsf{T}}\mathsf{P}\,\Delta\beta \;=\; \frac{(\beta_1-\beta_2)^2}{2s^2(1-r)}. \tag{36}$$

The projected prior charges only for the *difference* of the two amplitudes. A uniform shift, $\beta_1 = \beta_2$, is now free, as in the tied model; a shape change keeps exactly the penalty of Eq. (35).

The denominator is not an arbitrary constant. With $\mathrm{Var}(\beta_i) = s^2$ and $\mathrm{Cov}(\beta_1,\beta_2) = s^2 r$,

$$\mathrm{Var}(\beta_1-\beta_2) \;=\; s^2 + s^2 - 2s^2 r \;=\; 2s^2(1-r)\,: \tag{37}$$

the projected penalty is the squared jump between the bins divided by the prior variance of that jump. The whole construction reduces to one sentence: *the prior constrains the jump, not the level.*

I Why the projection "sees" the uniform shift: three views

Equation (14) subtracts from C^{-1} a matrix built out of the single vector $v = \mathsf{C}^{-1}\mathbf{1}$: the subtracted piece is $v\,v^{\mathsf{T}}$ divided by the number $\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}$. A matrix built from one vector acts along one direction only, so the subtraction removes the prior's information about exactly one direction, the uniform shift, and leaves everything else alone. This section derives the formula three times, each derivation adding its own intuition. Throughout, a group has n bins; $\Delta\beta$ and $\mathbf{1}$ are vectors with n entries.

I.1 View (a): subtracting the best-fitting uniform shift

Ask the following question: given a displacement $\Delta\beta$, how much of it can be explained as a pure change of the group amplitude? A change of B_g by c moves every β_i by c, so the candidate explanation is $c\,\mathbf{1}$, and the unexplained part should be judged by the prior metric. Minimize over the scalar c the quadratic form

$$\chi^2(c) \;=\; \big(\Delta\beta - c\,\mathbf{1}\big)^{\mathsf{T}}\mathsf{C}^{-1}\big(\Delta\beta - c\,\mathbf{1}\big). \tag{38}$$

Differentiating with respect to c (the two cross terms are equal because C^{-1} is symmetric),

$$\frac{d\chi^2}{dc} \;=\; -2\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\big(\Delta\beta - c\,\mathbf{1}\big) \;=\; 0, \qquad \frac{d^2\chi^2}{dc^2} \;=\; 2\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1} \;>\; 0, \tag{39}$$

where the second derivative is positive because C^{-1} is positive definite, so the solution of the first equation is the minimum of $\chi^2(c)$. That solution is

$$a \;=\; \frac{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\Delta\beta}{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}}. \tag{40}$$

For a diagonal C with variances s_i^2 this reduces to $a = \sum_i \Delta\beta_i/s_i^2 \big/ \sum_i 1/s_i^2$, the familiar inverse-variance weighted mean; Eq. (40) is its generalization to correlated errors: generalized least squares for a constant.

Keep the dimensions straight: $\Delta\beta$, $\mathbf{1}$ and the residual $w = \Delta\beta - a\,\mathbf{1}$ are vectors with n entries; a is a single number. The residuals w satisfy one linear constraint (Eq. (42) below), so they span $n-1$ dimensions: one direction of the n-dimensional displacement space has been handed to the amplitude, and $n-1$ directions of genuine shape freedom remain.

I.2 View (b): a Pythagorean split of the penalty

Write the displacement as the best uniform shift plus a residual,

$$\Delta\beta \;=\; a\,\mathbf{1} + w. \tag{41}$$

The first-order condition of Eq. (39) says precisely that

$$\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}w \;=\; 0\,: \tag{42}$$

the residual is orthogonal to the uniform direction *in the* C^{-1} *metric*. (With correlated errors, "orthogonal" must be measured with the same matrix that measures lengths.) Substituting the split into the full penalty, the cross term is $2a\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}w = 0$, and

$$\Delta\beta^{\mathsf{T}}\mathsf{C}^{-1}\Delta\beta \;=\; a^2\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1} \;+\; w^{\mathsf{T}}\mathsf{C}^{-1}w. \tag{43}$$

This is the Pythagorean theorem in the prior metric: the squared length of the displacement is the squared length of its uniform piece plus the squared length of its shape piece.

Now compare with the projection. Inserting Eq. (40) into the subtracted term of Eq. (14) shows that the subtraction removes exactly $a^2\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}$, the uniform piece. What remains is

$$\Delta\beta^{\mathsf{T}}\mathsf{P}\,\Delta\beta \;=\; w^{\mathsf{T}}\mathsf{C}^{-1}w. \tag{44}$$

Two consequences follow immediately. First, a pure uniform shift has $w = 0$, so its penalty vanishes. Equivalently $\mathsf{P}\,\mathbf{1} = 0$: the projected matrix sends the uniform direction to zero and charges nothing for it. Second, a pure shape change (a displacement that already satisfies Eq. (42)) has $a = 0$ and keeps its original penalty untouched. The projection removes one direction and alters nothing else.

I.3 View (c): marginalizing a flat-prior amplitude

The third view explains why the projection, rather than some other uniform-shift repair, is the *correct* prior. Model the group amplitudes as

$$\beta \;=\; B\,\mathbf{1} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \mathsf{C}), \tag{45}$$

with a flat prior on the scalar B. This is literally the intermediate model of Eq. (4), linearized around the fiducial. The prior density of β is obtained by integrating B out:

$$p(\beta) \;\propto\; \int dB\, \exp\!\Big[-\tfrac{1}{2}\big(\beta - B\,\mathbf{1}\big)^{\mathsf{T}}\mathsf{C}^{-1}\big(\beta - B\,\mathbf{1}\big)\Big]. \tag{46}$$

The exponent is quadratic in B. Writing $S = \mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}$ and completing the square,

$$\big(\beta - B\mathbf{1}\big)^{\mathsf{T}}\mathsf{C}^{-1}\big(\beta - B\mathbf{1}\big) \;=\; S\,\big(B - B_*\big)^2 \;+\; \beta^{\mathsf{T}}\mathsf{C}^{-1}\beta \;-\; \frac{\big(\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\beta\big)^2}{S}, \qquad B_* = \frac{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\beta}{S}. \tag{47}$$

The B integral is a Gaussian centered on B_* with width $1/\sqrt{S}$; it contributes only a β-independent constant $\sqrt{2\pi/S}$. What survives in the exponent is

$$p(\beta) \;\propto\; \exp\Big[-\tfrac{1}{2}\,\beta^{\mathsf{T}}\mathsf{P}\,\beta\Big], \qquad \mathsf{P} \;=\; \mathsf{C}^{-1} - \frac{\mathsf{C}^{-1}\mathbf{1}\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}}{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}}, \tag{48}$$

exactly the projection of Eq. (14).

The same computation has a standard matrix name. Expanding the exponent of Eq. (46) in the joint variables (β, B) gives a Gaussian with the $(n+1)\times(n+1)$ block precision

$$\mathsf{F}_{(\beta,B)} \;=\; \begin{pmatrix} \mathsf{C}^{-1} & -\,\mathsf{C}^{-1}\mathbf{1} \\ -\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1} & \mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1} \end{pmatrix}. \tag{49}$$

Removing a parameter from a precision matrix is not just deleting its row and column: the remaining block must also be corrected downward by whatever the removed parameter could absorb. The corrected block, $\mathsf{P} = \mathsf{F}_{\beta\beta} - \mathsf{F}_{\beta B}\,\mathsf{F}_{BB}^{-1}\,\mathsf{F}_{B\beta}$, is called the *Schur complement* over B, and it reproduces Eq. (14) term by term. So P is not a trick: it is precisely the statement "*B_g is sampled with a flat prior and marginalized*", folded into one matrix.

Note also that B_* in Eq. (47) is the same generalized-least-squares mean as a in Eq. (40): the marginalization integrates over all amplitudes, and the best-fitting one dominates the integral.

I.4 Checking views (b) and (c) on the two-bin case

For $n = 2$ the rows of C^{-1} in Eq. (33) have equal sums, so Eq. (40) gives the plain mean, $a = (\Delta\beta_1 + \Delta\beta_2)/2$, and the residual is the antisymmetric half-difference, $w = \frac{1}{2}(\Delta\beta_1 - \Delta\beta_2)\,(1,-1)^{\mathsf{T}}$. Its penalty, using the shape-change result of Eq. (35) with $a \to \frac{1}{2}(\Delta\beta_1 - \Delta\beta_2)$, is

$$w^{\mathsf{T}}\mathsf{C}^{-1}w \;=\; \frac{2}{s^2(1-r)}\left(\frac{\Delta\beta_1 - \Delta\beta_2}{2}\right)^2 \;=\; \frac{(\Delta\beta_1 - \Delta\beta_2)^2}{2s^2(1-r)}, \tag{50}$$

which is the direct two-bin computation of Eq. (36). Views (a), (b) and (c) all land on the same matrix.

J Many bins and several groups

J.1 Three bins: total correlation versus direct links

A group now has n bins at redshifts $z_1 < z_2 < \cdots < z_n$, unevenly spaced in general, with the prior covariance of Eq. (7),

$$\mathsf{C}_{ij} \;=\; s^2 \exp\!\left(-\,\frac{|z_i - z_j|}{L}\right). \tag{51}$$

This matrix is positive definite for any set of distinct redshifts (all its eigenvalues are positive), so the prior is a proper probability distribution and C^{-1} exists.

The structure of C^{-1} is best seen on three evenly spaced bins. With neighbor correlation $r = e^{-\Delta z/L}$, the outer pair sits two steps apart and carries $e^{-2\Delta z/L} = r^2$, so

$$\mathsf{C} \;=\; s^2\begin{pmatrix} 1 & r & r^2 \\ r & 1 & r \\ r^2 & r & 1 \end{pmatrix}, \qquad \mathsf{C}^{-1} \;=\; \frac{1}{s^2\,(1-r^2)}\begin{pmatrix} 1 & -r & 0 \\ -r & 1+r^2 & -r \\ 0 & -r & 1 \end{pmatrix}. \tag{52}$$

One multiplication confirms the surprising entry. The $(1,3)$ entry of $\mathsf{C}^{-1}\mathsf{C}$ must be zero (it is an off-diagonal entry of the identity matrix), and indeed row 1 of C^{-1} times column 3 of C gives, dropping the prefactors,

$$1 \cdot r^2 \;+\; (-r) \cdot r \;+\; 0 \cdot 1 \;=\; r^2 - r^2 \;=\; 0. \tag{53}$$

Here is the point that looks paradoxical at first. Bins 1 and 3 *are* correlated (C contains r^2), yet their entry in C^{-1} is exactly zero. The two matrices answer different questions. C gives the *total* correlation: what δ_1 tells you about δ_3 when nothing else is known. C^{-1} gives the *direct* correlation: what δ_1 still tells you about δ_3 once δ_2 is already known, and that is nothing. All of the 1–3 correlation travels through bin 2.

Think of a chain of people passing a message. The first person's message reaches far down the chain, but the links are only between neighbors: nobody talks past the person standing next to them.

The exponential kernel is special in making this exact, because its correlations multiply along the chain:

$$e^{-(z_3-z_1)/L} \;=\; e^{-(z_3-z_2)/L}\, e^{-(z_2-z_1)/L}. \tag{54}$$

The long-distance correlation is the product of the step correlations, so the middle bin already carries everything the outer bins share. A Gaussian kernel $e^{-\Delta z^2/L^2}$ does not factorize this way, because the squared separations do not add up along the chain: $(z_3-z_1)^2 \neq (z_3-z_2)^2+(z_2-z_1)^2$. Its inverse is dense.

The same structure shows directly in the penalty. For n evenly spaced bins, the quadratic form of the deviations δ_i (whose covariance is $\sigma_b^2\mathsf{R}$) regroups into a sum over neighbor links,

$$\delta^{\mathsf{T}}\big(\sigma_b^2\mathsf{R}\big)^{-1}\delta \;=\; \frac{\delta_1^2}{\sigma_b^2} \;+\; \sum_{k=1}^{n-1} \frac{\big(\delta_{k+1} - r\,\delta_k\big)^2}{\sigma_b^2\,(1-r^2)}. \tag{55}$$

Each term compares one bin with r times the bin before it: every bin is pulled only by its neighbor, and the long-range correlation comes from chaining the pulls. (Expanding the squares and collecting terms reproduces Eq. (52) for $n = 3$; Appendix K does it for general, uneven spacing.) This is the precise sense of "correlation only between adjacent bins": *directly*, only neighbors talk; *in total*, the influence propagates down the whole chain. Probability theory calls this neighbors-only structure the Markov property.

J.2 The tridiagonal inverse for n bins

A matrix whose nonzero entries sit only on the diagonal and the two neighboring off-diagonals is called *tridiagonal*. The three-bin inverse of Eq. (52) is the smallest interesting example, and the pattern grows unchanged. For n evenly spaced bins, where every adjacent pair has the same correlation $r = e^{-\Delta z/L}$ and $\mathsf{C}_{ij} = s^2\, r^{|i-j|}$,

$$\mathsf{C}^{-1} \;=\; \frac{1}{s^2\,(1-r^2)} \begin{pmatrix} 1 & -r & & & \\ -r & 1+r^2 & -r & & \\ & -r & \ddots & \ddots & \\ & & \ddots & 1+r^2 & -r \\ & & & -r & 1 \end{pmatrix}, \tag{56}$$

with diagonal $(1,\ 1+r^2,\ \ldots,\ 1+r^2,\ 1)$, off-diagonal entries $-r$, blanks zero, and the overall factor $1/\big[s^2(1-r^2)\big]$. The entries read off the chain sum of Eq. (55). An off-diagonal entry is the link between that pair of neighbors. An interior bin touches two links, one to the bin on its left and one to the bin on its right, and the two pulls together contribute $1+r^2$; the first and last bins touch only one link each, hence the plain 1 in the corners.

With uneven spacing, each adjacent pair has its own step and therefore its own correlation,

$$r_k \;=\; \exp\!\left(-\,\frac{z_{k+1}-z_k}{L}\right), \qquad k=1,\ldots,n-1, \tag{57}$$

and the inverse keeps the same shape with per-pair entries:

$$s^2\left(\mathsf{C}^{-1}\right)_{kk} = \begin{cases} \dfrac{1}{1-r_1^2}, & k=1, \\ \dfrac{1}{1-r_{k-1}^2}+\dfrac{r_k^2}{1-r_k^2}, & 1<k<n, \\ \dfrac{1}{1-r_{n-1}^2}, & k=n, \end{cases} \qquad s^2\left(\mathsf{C}^{-1}\right)_{k,k+1} = \frac{-\,r_k}{1-r_k^2}. \tag{58}$$

In words: the off-diagonal entry is the neighbor link of pair k, stronger when the bins are closer (r_k nearer one); the diagonal entry of bin k collects the links touching that bin, one term from the pair on its left and one from the pair on its right; the first and last bins carry one link only. Setting all $r_k = r$ turns the interior diagonal into $1/(1-r^2)+r^2/(1-r^2) = (1+r^2)/(1-r^2)$ and recovers Eq. (56). Appendix K derives Eq. (58) from the chain of neighbor links; the companion script `verify_prior.py` checks the three-bin, even and uneven forms numerically.

J.3 The projection, group by group

Each group receives the projected precision of Eq. (14), built from its own C. The properties derived in Appendix I carry over verbatim to n bins. $\mathsf{P}\,\mathbf{1} = 0$: a uniform shift of the whole group costs nothing. P is positive semidefinite, which means no displacement is *rewarded*: every move either costs χ^2 or is free. And of the n directions a group can move in, exactly one (the uniform shift) is free while the other $n-1$ are charged: in matrix language, P has rank $n-1$.

Several groups combine trivially. The prior of Eq. (5) is zero across groups, so the full prior covariance is block-diagonal, one block per group; and each group has its own free amplitude B_g, so each block is projected separately. The total prior precision is the block-diagonal matrix of the per-group projections.

J.4 The two limits

The intermediate model contains both extremes.

As $L \to \infty$, every $r_k \to 1$, so the jump variance $2\sigma_b^2(1-r_k)$ of every adjacent pair goes to zero: the prior forces all deviations of a group to be equal. The only freedom left is their common value, a uniform shift, and that is the direction the projection leaves free. This is the tied model, with one free B_g per group, *whatever* the value of σ_b.

As $\sigma_b \to \infty$ at fixed L, the overall scale of C diverges, so $\mathsf{C}^{-1} \to 0$ and with it $\mathsf{P} \to 0$: the prior penalty vanishes and every bin's amplitude is free: the independent model with flat priors.

Between the limits, L and σ_b interpolate continuously, which is what makes the model useful: the figure of merit can be traced as a function of how much bin-to-bin freedom the bias is granted.

K The tridiagonal inverse from the chain of links

Because correlations multiply along the chain (Eq. (54)), the prior can be built one bin at a time: the joint density of the deviations factorizes into neighbor conditionals,

$$p(\delta_1, \ldots, \delta_n) \;=\; p(\delta_1)\, p(\delta_2 \mid \delta_1) \cdots p(\delta_n \mid \delta_{n-1}), \tag{59}$$

where each conditional is Gaussian with

$$\big\langle \delta_{k+1} \mid \delta_k \big\rangle \;=\; r_k\, \delta_k, \qquad \mathrm{Var}\big(\delta_{k+1} \mid \delta_k\big) \;=\; \sigma_b^2 \left(1 - r_k^2\right). \tag{60}$$

Here $r_k = e^{-(z_{k+1}-z_k)/L}$ is, as in Appendices B.2 and J.2, the correlation coefficient of the adjacent bins k and $k+1$: the kernel of Eq. (5) gives $\mathrm{Cov}(\delta_k, \delta_{k+1}) = \sigma_b^2\, r_k$ and $\mathrm{Var}(\delta_k) = \mathrm{Var}(\delta_{k+1}) = \sigma_b^2$.

The conditional mean and variance in Eq. (60) follow from nothing more than these covariances. Split δ_{k+1} into the part predicted by its neighbor and a leftover e,

$$\delta_{k+1} \;=\; a\, \delta_k + e, \tag{61}$$

and choose the number a so that the leftover carries no correlation with δ_k:

$$\mathrm{Cov}(e, \delta_k) \;=\; \mathrm{Cov}(\delta_{k+1}, \delta_k) - a\, \mathrm{Var}(\delta_k) \;=\; \sigma_b^2\, r_k - a\, \sigma_b^2 \;=\; 0 \quad\Longrightarrow\quad a = r_k. \tag{62}$$

The leftover's variance follows by expanding $e = \delta_{k+1} - r_k \delta_k$:

$$\mathrm{Var}(e) \;=\; \mathrm{Var}(\delta_{k+1}) - 2 r_k\, \mathrm{Cov}(\delta_{k+1}, \delta_k) + r_k^2\, \mathrm{Var}(\delta_k) \;=\; \sigma_b^2 - 2 r_k^2 \sigma_b^2 + r_k^2 \sigma_b^2 \;=\; \sigma_b^2 \left(1 - r_k^2\right). \tag{63}$$

For jointly Gaussian variables, uncorrelated means independent. So once δ_k is known, the only randomness left in $\delta_{k+1} = r_k \delta_k + e$ is e itself: the conditional mean is $r_k \delta_k$ and the conditional variance is $\sigma_b^2(1 - r_k^2)$, which is Eq. (60). In one line: knowing the neighbor explains the fraction r_k^2 of the bin's variance, and the fraction $1 - r_k^2$ is new.

The factorization of Eq. (59) needs one more fact: conditioning on δ_k alone must be enough: the earlier bins must add nothing. This is again the multiplication of correlations along the chain. For any earlier bin $j \le k$,

$$\mathrm{Cov}(e, \delta_j) \;=\; \sigma_b^2 \left[e^{-(z_{k+1}-z_j)/L} - r_k\, e^{-(z_k - z_j)/L} \right] \;=\; \sigma_b^2\, e^{-(z_k - z_j)/L} \left[e^{-(z_{k+1}-z_k)/L} - r_k \right] \;=\; 0, \tag{64}$$

because the first exponential factorizes through z_k and the bracket is $r_k - r_k$. The leftover is therefore independent of every earlier bin, and the chain of one-step conditionals is the complete prior.

Taking $-2 \ln p$ of Eq. (59) gives the penalty as the chain sum, the uneven-spacing version of Eq. (55):

$$\delta^{\mathsf{T}} \big(\sigma_b^2 \mathsf{R}\big)^{-1} \delta \;=\; \frac{\delta_1^2}{\sigma_b^2} \;+\; \sum_{k=1}^{n-1} \frac{\big(\delta_{k+1} - r_k\, \delta_k\big)^2}{\sigma_b^2 \left(1 - r_k^2\right)}. \tag{65}$$

Expanding each square and collecting the coefficient of every product $\delta_i\delta_j$ reads off the precision matrix entry by entry. δ_1^2 appears twice: in the first term (coefficient 1) and in the $k=1$ link as the predictor (coefficient $r_1^2/(1-r_1^2)$); the two add up to $1/(1-r_1^2)$. An interior δ_k^2 appears in the $k-1$ link as the predicted bin ($1/(1-r_{k-1}^2)$) and in the k link as the predictor ($r_k^2/(1-r_k^2)$). The last bin is only ever predicted: $1/(1-r_{n-1}^2)$. Each product $\delta_k\delta_{k+1}$ appears in its own link only, with coefficient $-2r_k/(1-r_k^2)$, i.e. a symmetric matrix entry $-r_k/(1-r_k^2)$. No product of non-neighbors ever appears, so the matrix is tridiagonal by construction. This is Eq. (58) in units of σ_b^2; replacing δ by $\Delta\beta = B_g\,\delta$ and σ_b by $s = \sigma_b B_g$ gives the same matrix for the bin amplitudes.

L The memory projection, step by step

Throughout, a group has n bins, C is the kernel covariance $\mathsf{C}_{ij} = \sigma_b^2 e^{-|z_i-z_j|/L}$ of the residuals, $\mathsf{Q} = \mathsf{C}^{-1}$, $\mathsf{U} = [\,\mathbf{1},\ v\,]$ is the $n\times 2$ design matrix of Eq. (23), and $\theta = (a, s_g)^{\mathsf{T}}$ collects a line's level and slope. (For the bin amplitudes β, every covariance carries a factor B_g^2 that cancels in each identity below.)

L.1 The line fit, the split, and the marginalization

The three views of Appendix I generalize from the single column $\mathbf{1}$ to U without change of structure.

Minimizing the penalty of a displacement δ over the line, $\chi^2(\theta) = (\delta - \mathsf{U}\theta)^{\mathsf{T}}\mathsf{Q}\,(\delta - \mathsf{U}\theta)$, gives the normal equations of a generalized least-squares fit,

$$\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\,\hat\theta \;=\; \mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\delta, \qquad \hat\theta = (\hat a, \hat s_g)^{\mathsf{T}}, \tag{66}$$

with $\mathsf{U}^{\mathsf{T}}\mathsf{Q}\mathsf{U}$ a 2×2 matrix, positive definite whenever the group has at least two distinct redshifts (then $\mathbf{1}$ and v are linearly independent and U has full column rank).

Writing $\delta = \mathsf{U}\hat\theta + w$, the normal equations say $\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,w = 0$: the residual is orthogonal to both columns in the Q metric. The cross terms then vanish and the penalty splits,

$$\delta^{\mathsf{T}}\mathsf{Q}\,\delta \;=\; \hat\theta^{\mathsf{T}}\big(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)\hat\theta \;+\; w^{\mathsf{T}}\mathsf{Q}\,w, \qquad \delta^{\mathsf{T}}\mathsf{P}_{\rm mem}\,\delta \;=\; w^{\mathsf{T}}\mathsf{Q}\,w\,: \tag{67}$$

the projection keeps the residual's penalty and discards the line's. A pure line has $w=0$, so $\mathsf{P}_{\rm mem}\mathsf{U} = 0$; the residuals obey two linear constraints, so $\mathsf{P}_{\rm mem}$ has rank $n-2$; and $w^{\mathsf{T}}\mathsf{Q}\,w \ge 0$ makes it positive semidefinite.

Marginalization gives the same matrix. With $\delta \sim \mathcal{N}(\mathsf{U}\theta, \mathsf{C})$ and $\theta \sim \mathcal{N}(0,\Lambda)$, $\Lambda = \mathrm{diag}(W^2,\tau^2)$, the marginal covariance of δ is $\mathsf{C} + \mathsf{U}\Lambda\mathsf{U}^{\mathsf{T}}$, and the Woodbury identity inverts it:

$$\big(\mathsf{C} + \mathsf{U}\Lambda\mathsf{U}^{\mathsf{T}}\big)^{-1} \;=\; \mathsf{Q} - \mathsf{Q}\,\mathsf{U}\big(\Lambda^{-1} + \mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)^{-1}\mathsf{U}^{\mathsf{T}}\mathsf{Q}. \tag{68}$$

As the prior widths grow, $\Lambda^{-1}\to 0$ and the right-hand side becomes $\mathsf{P}_{\rm mem}$. The deviation falls inverse-quadratically in the widths: expanding the middle inverse, the leading correction carries Λ^{-1}, so the deviation falls as $1/W^2$, a factor of one hundred per decade of W. The script of Appendix Q verifies the fall.

L.2 The null-space form

A second form of the projection makes its meaning plain. Let N be an $n \times (n-2)$ matrix whose columns span the combinations of bins that no straight line can produce: $\mathsf{N}^{\mathsf{T}}\mathsf{U} = 0$, plain Euclidean orthogonality to $\mathbf{1}$ and to v. Then

$$\mathsf{Q} - \mathsf{Q}\,\mathsf{U}\big(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)^{-1}\mathsf{U}^{\mathsf{T}}\mathsf{Q} \;=\; \mathsf{N}\big(\mathsf{N}^{\mathsf{T}}\mathsf{C}\,\mathsf{N}\big)^{-1}\mathsf{N}^{\mathsf{T}}. \tag{69}$$

The proof has three steps. Both sides annihilate U: the left by construction, the right because $\mathsf{N}^{\mathsf{T}}\mathsf{U} = 0$. Both sides map CN to N: the right because $\mathsf{N}(\mathsf{N}^{\mathsf{T}}\mathsf{CN})^{-1}(\mathsf{N}^{\mathsf{T}}\mathsf{CN}) = \mathsf{N}$, the left because $\mathsf{QCN} = \mathsf{N}$ and the subtracted term meets $\mathsf{U}^{\mathsf{T}}\mathsf{N} = 0$. And the columns of CN and of U together span the whole space: if $\mathsf{CN}x + \mathsf{U}y = 0$, multiplying by N^{T} gives $\mathsf{N}^{\mathsf{T}}\mathsf{CN}\,x = 0$, so $x = 0$ (the middle matrix is positive definite), and then $y = 0$ (U has full column rank). Two matrices that agree on a spanning set are equal.

The meaning: since $\delta^{\mathsf{T}}\mathsf{P}_{\text{mem}}\,\delta = (\mathsf{N}^{\mathsf{T}}\delta)^{\mathsf{T}}(\mathsf{N}^{\mathsf{T}}\mathsf{C}\,\mathsf{N})^{-1}(\mathsf{N}^{\mathsf{T}}\delta)$, the prior reads off the combinations $\mathsf{N}^{\mathsf{T}}\delta$ that no line can make and charges each by the inverse of its own prior covariance. The same identity with the single column $\mathsf{U} = \mathbf{1}$ gives the kernel-model projection of Eq. (14); there N spans the zero-sum combinations, the differences between bins.

L.3 Worked examples

One or two bins are fully free. For $n = 2$ the matrix U is square and invertible (two distinct redshifts), so $(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\mathsf{U})^{-1} = \mathsf{U}^{-1}\mathsf{C}\,(\mathsf{U}^{\mathsf{T}})^{-1}$ and the subtracted term collapses to Q: the projection vanishes. For $n = 1$ nothing is orthogonal to both columns, so N has no columns and the null-space form is the zero matrix: a line fixes a single value trivially.

Three evenly spaced bins show what survives. With spacing Δ and $r = e^{-\Delta/L}$, the centered redshifts are $v = (-\Delta, 0, \Delta)^{\mathsf{T}}$, and the only direction orthogonal to $\mathbf{1}$ and v is the discrete curvature, the second difference $\mathsf{N} = (1, -2, 1)^{\mathsf{T}}$. Its prior variance expands to

$$\mathsf{N}^{\mathsf{T}}\mathsf{C}\,\mathsf{N} \;=\; \sigma_b^2\,\big(6 - 8r + 2r^2\big) \;=\; 2\,\sigma_b^2\,(1-r)(3-r) \tag{70}$$

(the diagonal contributes $1+4+1 = 6$, the two nearest-neighbor pairs $-8r$, and the outer pair $+2r^2$), so the penalty is

$$\delta^{\mathsf{T}}\mathsf{P}_{\text{mem}}\,\delta \;=\; \frac{\big(\delta_1 - 2\delta_2 + \delta_3\big)^2}{2\,\sigma_b^2\,(1-r)(3-r)}. \tag{71}$$

The memory model charges only the curvature; it is blind to any straight-line drift. The kernel model's three-bin case (Appendix J.1) charges the two neighbor differences instead, so it taxes a drift as heavily as a zigzag of the same step size.

L.4 The universal-kriging predictor

Predict bin $k+1$ from the first k bins, $k \geq 2$, with flat priors on θ. Write C_k for the past kernel block, U_k for the past rows of U, $u_{k+1} = (1, v_{k+1})^{\mathsf{T}}$ for the next row, and c for the vector of kernel covariances between bin $k+1$ and the past. Given θ, the ordinary Gaussian conditional is

$$\mathrm{E}\big[\delta_{k+1} \mid \delta_{1..k}, \theta\big] \;=\; u_{k+1}^{\mathsf{T}}\theta \;+\; c^{\mathsf{T}}\mathsf{C}_k^{-1}\big(\delta_{1..k} - \mathsf{U}_k\,\theta\big). \tag{72}$$

With a flat prior, the posterior of θ given the past is centered on the generalized-least-squares fit $\hat{\theta}$ of Eq. (66) applied to the past block. Equation (72) is linear in θ, so averaging over the posterior replaces θ by $\hat{\theta}$.

For the exponential kernel the weights collapse. Correlations multiply along the chain (Eq. (54)), so every entry of c is r_k times the corresponding entry of C_k's last column, and

$$c^{\mathsf{T}}\mathsf{C}_k^{-1} \;=\; r_k\, e_k^{\mathsf{T}}, \tag{73}$$

with e_k the k-th unit vector (one in the last past slot, zero elsewhere). Substituting both results into Eq. (72) gives Eq. (22): the fitted line evaluated at the new bin, plus one Markov step from the last residual.

L.5 The finite-slope correlation

With a Gaussian slope of variance τ^2 and the amplitude held in B_g, the deviations have the covariance of Eq. (20), so $\mathrm{Var}(\delta_i) = \tau^2 v_i^2 + \sigma_b^2$ and $\mathrm{Cov}(\delta_i, \delta_j) = \tau^2 v_i v_j + \sigma_b^2 e^{-|z_i - z_j|/L}$. For separations much larger than L the kernel term is exponentially small, and dropping it gives the correlation coefficient of Eq. (21). When $\tau|v_i|$ and $\tau|v_j|$ both exceed σ_b, the σ_b^2 terms in the denominators become negligible and the coefficient tends to $v_i v_j/|v_i v_j| = \pm 1$: plus one for bins on the same side of the group center, minus one for opposite sides.

M The bias continuity equation, step by step

The matter density $\rho_m(\mathbf{x}, t)$ defines the matter overdensity $\delta_m = (\rho_m - \bar{\rho}_m)/\bar{\rho}_m$, with $\bar{\rho}_m(t)$ the mean density; the galaxy number density n_g defines the galaxy-count overdensity δ_g the same way. Conservation of mass and of galaxy number gives each field a continuity equation. In comoving coordinates, keeping only terms linear in the fluctuations (the convective term $\mathbf{v} \cdot \nabla\delta$ is second order and dropped),

$$\frac{\partial \delta_m}{\partial t} \;=\; -\,\frac{1}{a}\,\nabla \cdot \mathbf{v}_m, \qquad \frac{\partial \delta_g}{\partial t} \;=\; -\,\frac{1}{a}\,\nabla \cdot \mathbf{v}_g, \tag{74}$$

where t is cosmic time, a the scale factor, and $\mathbf{v}_m$ and $\mathbf{v}_g$ the peculiar velocity fields of the matter and of the galaxies.

The two velocity fields are the same. On large scales the galaxies carry no pressure of their own and fall in the same gravitational potential as the matter, so a conserved sample rides the matter flow: $\mathbf{v}_g = \mathbf{v}_m$. This is the no-velocity-bias assumption; Tegmark and Peebles [2] state it as galaxies behaving as test particles that fairly trace the large-scale velocity field.

With equal right-hand sides, subtracting the two equations of Eq. (74) leaves

$$\frac{\partial}{\partial t}\big(\delta_g - \delta_m\big) \;=\; 0 \quad\Longrightarrow\quad \delta_g(z) - \delta_m(z) \;=\; \delta_g(z_c) - \delta_m(z_c) \tag{75}$$

after integrating in time from the reference redshift z_c to any other redshift z: the excess clumping keeps its initial value forever.

Now substitute the bias. At z_c the sample has linear bias b_c, so $\delta_g(z_c) = b_c\,\delta_m(z_c)$ and the frozen difference is $(b_c - 1)\,\delta_m(z_c)$. The matter grows as $\delta_m(z) = \delta_m(z_c)\,D(z)/D(z_c)$, with D the linear growth factor. Then

$$\delta_g(z) \;=\; \delta_m(z) + \big(b_c - 1\big)\,\delta_m(z_c) \;=\; \delta_m(z)\left[\,1 + \big(b_c - 1\big)\,\frac{D(z_c)}{D(z)}\,\right], \tag{76}$$

and dividing by $\delta_m(z)$ gives the bias $b(z) = \delta_g/\delta_m$ of Eq. (15). Dividing instead by the tied bias $b_c\, D(z_c)/D(z)$ gives the deviation of Eq. (16), $\delta(z) = (D(z)/D(z_c) - 1)/b_c$: positive at redshifts below z_c, where the matter has grown past its reference value, and negative above.

The debiasing limit follows from Eq. (15) alone. As D grows, the frozen term is divided by an ever larger number, so $b \to 1$: from above when $b_c > 1$ and from below when $b_c < 1$. Gravity debiases a conserved sample, because the part of δ_g that traces the matter keeps growing while the frozen excess does not.

Tegmark and Peebles drop the assumption that δ_g is proportional to δ_m point by point. They describe the pair of fields by three numbers: the matter fluctuation amplitude σ, the bias b (the ratio of the galaxy to the matter fluctuation amplitude) and the correlation coefficient ρ_{gm} between the two fields; $\rho_{gm} < 1$ is stochastic bias, the galaxy field tracing the matter imperfectly. (The subscript keeps it apart from the neighbor correlation r of Eq. (9).) The same continuity argument evolves all three (their section "After the galaxy formation epoch"): with D normalized to one today and the subscript zero marking today,

$$b_0 = \left[(1-D)^2 + 2D(1-D)\, b\, \rho_{gm} + D^2 b^2\right]^{1/2}, \qquad \rho_{gm,0} = \frac{(1-D) + D\, b\, \rho_{gm}}{b_0}, \tag{77}$$

where b and ρ_{gm} are the values at the epoch whose growth factor is D.

Both b_0 and $\rho_{gm,0}$ approach one as clustering grows, whatever their initial values: the matter-sourced part of the galaxy field grows with D while the frozen part does not, so the field is progressively dominated by the piece that traces the matter. Perfect correlation is a fixed point: for $\rho_{gm} = 1$, Eq. (77) collapses to $b_0 - 1 = (b-1)\, D$, which is Eq. (15) again. The conserved population of this note is the perfectly correlated special case of their model.

Each assumption fails in a specific way for a DESI sample. Membership is not conserved: the flux limit and color cuts admit and remove galaxies as redshift grows (the threshold drift and color-selection features of Appendix C.4), mergers turn two members into one, and an ELG belongs to the sample only during its star-forming phase. The no-velocity-bias and linear-theory assumptions hold well on the scales this forecast keeps; conservation is the assumption that breaks.

The script of Appendix Q integrates Eq. (74) numerically, with δ_m from the linear growth equation, and recovers the closed form of Eq. (15) to a relative accuracy of 2×10^{-9}.

N The BGS flux-limit bias model, step by step

Appendix E.4 quotes a BGS bias drift driven by the survey's flux limit: a 22.3% span of the deviation, a convex shape, and a 2.4% rms residual around the best straight line. This appendix derives that model in four steps, tabulates every intermediate quantity, explains the shape, and then lists what the model leaves out. The model lives in the function `bgs_flux_limit_bias` of `plots/make_theory_data.py`, and the script of Appendix Q checks the numbers below.

N.1 The four steps

Step 1: the flux limit becomes a luminosity threshold. BGS Bright keeps every galaxy brighter than apparent magnitude $r = 19.5$ (measured limit, [15]). Apparent magnitude m and absolute magnitude M are related by $m = M + 25 + 5\log_{10}[d_L/(h^{-1}\,\mathrm{Mpc})] + K(z)$, where d_L is the luminosity distance, h is the Hubble constant in units of 100 km/s/Mpc, and $K(z)$

is the K-correction: the magnitude shift that accounts for a fixed filter sampling a redshifted spectrum [14]. The model sets $K = 0$. The faintest absolute magnitude that passes the cut at redshift z is then

$$M_{\rm lim}(z) - 5\log_{10} h \;=\; 19.5 - 25 - 5\log_{10}\frac{d_L(z)}{h^{-1}\,{\rm Mpc}}, \qquad d_L(z) \;=\; (1+z)\,\chi(z), \tag{78}$$

with $\chi(z)$ the comoving distance of flat ΛCDM with $\Omega_m = 0.3$, the background of the whole note. Every magnitude in this appendix carries the $M - 5\log_{10} h$ convention, the convention of the M_* in step 3. Brighter means more negative: between the first and last $\Delta z = 0.05$ bin centers the threshold brightens by 2.68 magnitudes (Table 10).

Step 2: the sample at each redshift is a luminosity-threshold sample. At redshift z the model sample holds every galaxy brighter than the threshold, $M < M_{\rm lim}(z)$. Each thin bin is therefore a luminosity-threshold sample, and the threshold brightens with distance: the one selection function $r < 19.5$ turns into a different threshold sample in every bin.

Step 3: the threshold sets the bias through the Zehavi et al. fit. Zehavi et al. [12] measured the large-scale bias of luminosity-threshold samples in the SDSS main sample at $z < 0.25$ (effective redshift about 0.1) and fit it with

$$b(>L) \;=\; 1.06 + 0.21\,(L/L_*)^{1.12}, \qquad L/L_* \;=\; 10^{-0.4\,(M_{\rm lim} - M_*)}, \tag{79}$$

where L is the threshold luminosity of the sample, and L_* and $M_* - 5\log_{10} h = -20.44$ are the characteristic luminosity and magnitude of the SDSS r-band luminosity function [13]. Appendix O summarizes the measurement itself. The model applies the fit at every redshift of the group, without evolution; it also reads the published left side $b\,(\sigma_8/0.8)$ as b, with σ_8 the rms matter fluctuation in spheres of $8\,{\rm Mpc}/h$.

Step 4: the deviation from the tied shape. The deviation δ of Eq. (4) compares this bias with the tied shape $B_g/D(z)$, where D is the linear growth factor of Appendix A. Because the group amplitude B_g is free, only the shape of $b(z)$ matters, and the model fixes the amplitude B by making the mean deviation over the group's bin centers vanish:

$$\delta(z_i) \;=\; \frac{b(z_i)\,D(z_i)}{B} - 1, \qquad B \;=\; \frac{1}{n}\sum_{j=1}^{n} b(z_j)\,D(z_j), \tag{80}$$

with $z_1, \ldots, z_n$ the bin centers of the group.

Table 10 evaluates the chain at the $\Delta z = 0.05$ bin centers of the BGS group, the same evaluation points as Table 4, Figure 10 and the numbers of Appendix E.4: the threshold runs from -18.56 to -21.24, the bias from 1.09 to 1.54, and the deviation spans 22.3%. On a coarser grid the endpoint values move; at $z = 0.11$ and $z = 0.39$, for example, the bias runs from 1.08 to 1.59.

N.2 Why the drift is convex

The two factors of $b\,D$ change at very different rates: across the bin centers of Table 10 the bias rises by 41% while D falls by 12%, so the deviation inherits the shape of the bias curve.

z	d_L [h^{-1} Mpc]	$M_{\rm lim} - 5\log_{10} h$	L/L_*	b	D	δ
0.125	410	−18.56	0.18	1.09	0.937	−8.0%
0.175	591	−19.36	0.37	1.13	0.913	−7.2%
0.225	783	−19.97	0.65	1.19	0.890	−4.7%
0.275	984	−20.46	1.02	1.27	0.867	−0.4%
0.325	1192	−20.88	1.50	1.39	0.845	+5.9%
0.375	1409	−21.24	2.10	1.54	0.824	+14.4%

Table 10: The BGS flux-limit chain at the $\Delta z = 0.05$ bin centers: the luminosity distance d_L (Eq. (78)), the threshold magnitude $M_{\rm lim}$, the threshold luminosity L/L_* and bias b (Eq. (79)), the growth factor D, and the deviation δ from the tied shape (Eq. (80)).

That curve is flat at low redshift and steep at high redshift for one reason: the luminosity ratio is exponential in the threshold magnitude, $L/L_* = 10^{-0.4\,(M_{\rm lim} - M_*)}$. Below L_* the term $0.21\,(L/L_*)^{1.12}$ is small against the constant 1.06 (0.03 at the first bin center), so the bias stays near the faint-end floor of the fit. The threshold crosses L_* near $z = 0.27$, and past the crossing the same term dominates the growth of b and rises exponentially with the threshold magnitude (0.48 at the last center). A curve that is flat on one side of a crossing and exponential on the other is convex across the group.

A straight line through a convex curve leaves a U-shaped residual. Quantitatively, the deviation at the six centers is almost exactly a parabola: adding a quadratic term cuts the residual from 2.4% to 0.1% rms. The fitted quadratic coefficient is 3.8 per unit redshift squared, and a pure parabola with that coefficient, sampled at the six centers with its best line removed, leaves residuals of +3.2% at the ends and −2.6% in the middle, rms 2.4%: the measured residuals $(+3.1, -0.5, -2.5, -2.6, -0.8, +3.3)$% of Figure 10 are this U shape. The 2.4% rms of Appendix E.4 is therefore the curvature of the drift, the one part of it a free tilt cannot absorb.

N.3 What the model leaves out

Each omission below states the direction in which it pushes the drift. None of them changes the qualitative shape: a smooth, monotonic, convex rise of the bias toward the far edge of the group.

The threshold-bias relation is frozen at $z \approx 0.1$. Equation (79) is a fit to clustering at $z \lesssim 0.25$, applied here up to $z = 0.4$. Two neglected evolutions pull in opposite directions. First, the bias at fixed luminosity grows with redshift: for a conserved population the continuity equation of Appendix M gives $b - 1 \propto 1/D$ (Eq. (15)), so a threshold sample at $z = 0.4$ is more biased than the $z \approx 0.1$ calibration reports. This steepens the drift. Second, galaxies were brighter in the past: Blanton et al. [13] measure the luminosity evolution of the SDSS r-band luminosity function as a brightening of M_* by $Q \approx 1$–2 magnitudes per unit redshift. A fixed threshold therefore sits lower relative to $L_*(z)$ at high redshift, which lowers L/L_* and the bias there. This flattens the drift. The flattening is a fraction of the effect, not all of it: $Q\,(z - 0.1)$ reaches 0.3–0.6 magnitudes at the last bin center, against the 2.68 magnitudes the threshold itself brightens across the group.

No K-correction. The model drops the $K(z)$ of step 1. Two pieces are dropped with it. The bandpass-shift part alone, the special case of observing a spectrum flat in wavelength through its own filter, is $2.5\log_{10}(1+z)$, which is 0.37 magnitudes at $z = 0.4$ [14]; a red spectrum, falling toward the blue, makes the r-band correction larger still. In addition, the M_* of step 3 is measured in the r band shifted to $z = 0.1$ [13], while Eq. (78) produces an observed-band magnitude; the K-correction is exactly the bridge between the two bands, and the model omits the bridge. A positive K makes the true threshold brighter at high redshift than Eq. (78) says, so the true L/L_* and bias are higher there: the omission steepens the drift.

Only $r < 19.5$ is modeled. BGS Bright also imposes a fiber-magnitude cut for redshift success, and BGS Faint, the color-selected extension at $19.5 < r < 20.175$, is absent from the model [15]. A sample that includes BGS Faint has a limit 0.675 magnitudes fainter at every redshift, which lowers L/L_* and the bias everywhere and moves the L_* crossing to higher redshift: a flatter drift. The fiber cut and the color selection change the galaxy mix in a way the threshold fit of Eq. (79) cannot follow, and this model assigns them no direction.

Scope. The model is an order-of-magnitude estimate. Its reliable content is the shape of the drift (smooth, monotonic, convex) and its size to a factor of order unity, and that is the content this note uses: Appendix E.4 sizes the memory model's residual scatter for BGS from the 2–3% curvature. It is not a prediction of the measured DESI BGS bias. The upgrade path replaces Eq. (79) with DESI's own measurement: the One-Percent Survey analysis of Prada et al. [16] fits the BGS clustering as a function of absolute magnitude and redshift with abundance matching, and reports the dependence of the large-scale bias on absolute magnitude; that relation, plus the measured K-corrections and luminosity evolution above [14, 13], would replace steps 1 and 3 with calibrated inputs.

O What the literature says, paper by paper

This appendix summarizes each measurement behind Appendix C.5, for a reader who has not read the papers. The tags are those of that section: measured, derived, assumed.

O.1 Passive-evolution tests of massive red galaxies

Wake et al. [3]. What was measured: the two-point correlation function of over 10,000 luminous red galaxies (LRGs) from SDSS and the 2SLAQ survey at $0.19 < z < 0.55$, matched in rest-frame color and luminosity, with stellar masses above $10^{11}\,h^{-1}M_\odot$. Method: the redshift-space and real-space autocorrelation of the matched samples, interpreted with a halo model. Result: no significant evolution of the correlation amplitude, but a strictly merger-free passive model is rejected; a merger rate of $7.5 \pm 2.3\%$ between $z = 0.55$ and $z = 0.19$, about 2.4% per Gyr, fits the data. Implication here: LRG deviations from conserved evolution are a few percent in number per Gyr (measured), and the induced bias change is smaller, because merging moves galaxies between halos of similar mass.

Tojeiro et al. [4]. What was measured: the merger rate and clustering evolution of 250,000 BOSS galaxies up to $z \approx 0.7$, connected to the SDSS-I/II LRGs at lower redshift. Method: each BOSS galaxy receives a likelihood of being a progenitor of the LRG sample, built from

the fossil record of the LRGs (star-formation and metallicity histories); the weighted sample's large-scale clustering is compared between the surveys. Result: the sample evolves by merging at $2\pm1.5\%$ per Gyr, and the weighted clustering evolution is consistent with dynamical passive evolution, the continuity-equation prediction. Implication here: the conserved shape holds to the percent-per-Gyr level for these galaxies (measured).

Guo et al. [5]. What was measured: the projected two-point correlation function of about 260,000 CMASS galaxies at $0.43 < z < 0.7$ (BOSS DR9). Method: subsamples of fixed comoving number density built with redshift-dependent magnitude thresholds, so that the same population is followed across redshift. Result: the clustering of the fixed-density samples is consistent with passive evolution, and red-sequence clustering at fixed luminosity shows no significant evolution. Implication here: the same (measured).

O.2 The DESI luminous red galaxies

Zhou et al. [6]. What was measured: the redshift-dependent clustering of 2.7 million DESI-like LRGs over 5655 square degrees at $0.4 < z < 0.9$, in photometric-redshift bins of width $\Delta z \approx 0.1$. Method: projected correlation functions with photometric redshifts of 0.02 precision, fit with a five-parameter halo occupation distribution (HOD). Result: the HOD parameters show little evolution except in the highest bin, and the bias follows a constant clustering amplitude; Table 11 lists the measured biases and their residuals around $b = 1.5/D$ (derived here): within $\pm2.3\%$ below $z = 0.83$, and $+6.4\pm6.4\%$ at $0.84 < z < 0.93$. Implication: LRG deviations at $\Delta z \approx 0.1$ spacing are at most a few percent below $z = 0.83$ (measured plus derived).

photometric bin	b	residual around $1.5/D$
0.41–0.50	1.89 ± 0.03	$-0.4\% \pm 1.6\%$
0.50–0.61	2.04 ± 0.02	$+2.3\% \pm 1.0\%$
0.61–0.72	2.07 ± 0.03	$-1.6\% \pm 1.4\%$
0.75–0.83	$2.24^{+0.07}_{-0.08}$	$+0.4\% \pm 3.4\%$
0.84–0.93	2.48 ± 0.15	$+6.4\% \pm 6.4\%$

Table 11: The DESI-like LRG bias of Zhou et al. (2021, arXiv:2001.06018, Results section; measured) and its residual around the constant clustering amplitude $b = 1.5/D$ (derived, with flat ΛCDM $\Omega_m = 0.3$ growth factors at the bin centers).

Zhou et al. [7]. What was measured: the DESI LRG target selection and its spectroscopic performance. Method: selection cuts in g, r, z and $W1$ (the WISE 3.4 micron band), validated on Survey Validation spectra. Result: a sliding color-magnitude cut holds the comoving density near $5\times10^{-4}\,(h/\mathrm{Mpc})^3$ at $0.4 < z < 0.8$; the overall faint limit is $z_{\mathrm{fiber}} < 21.6$, where z_{fiber} is the z-band magnitude expected within one DESI fiber; 98.9% of observed targets yield confident redshifts. Implication: inside LRG1 and LRG2 the selection gives the threshold no room to jump (measured); past $z \approx 0.8$ the fixed flux limit drives the LRG3 drift of Appendix C.5 (derived). The chain of that derivation: a fixed flux limit raises the luminosity threshold with the distance modulus, by 0.27 magnitudes per $\Delta z = 0.1$ at $z \approx 1$; the Zehavi slope below is 0.5–1.0 in bias per magnitude near LRG luminosities; their product, 0.13–0.27 in bias, is

6–11% of $b \approx 2.4$. The caveat is stronger than for BGS: the threshold-bias fit is an r-band relation measured at $z < 0.25$, applied here to a z-band fiber-magnitude selection at $z \approx 0.8$–1.1, outside its calibrated range, so the drift's direction and order of magnitude are what the argument establishes.

White et al. [9] and Sailer et al. [10]. What was measured: cosmological constraints from the cross-correlation of the photometric LRG sample of [8] with CMB lensing from Planck, and in [10] also ACT DR6, in four tomographic bins over $0.4 < z < 1$. Method: angular cross- and auto-spectra with one free bias parameter per bin. Result: smooth models fit all bins. Implication here: these analyses neither require nor exclude percent-level bias structure between thin bins; they bound nothing below their bin width of about 0.15 (measured context).

O.3 Bias versus luminosity threshold

Zehavi et al. [12]. What was measured: the luminosity and color dependence of galaxy clustering in the SDSS DR7 main sample, about 700,000 galaxies at $z < 0.25$. Method: projected correlation functions of volume-limited luminosity-threshold samples, interpreted with HOD modeling. Result: the large-scale bias of a threshold sample is fit by $b\,(\sigma_8/0.8) = 1.06 + 0.21\,(L/L_*)^{1.12}$, with L_* the characteristic luminosity of the r-band luminosity function and σ_8 the rms matter fluctuation in spheres of $8\,\mathrm{Mpc}/h$; the measured threshold biases rise from 1.40 ± 0.03 at $M_r < -21.0$ to 2.16 ± 0.05 at $M_r < -22.0$, a slope of 0.5–1.0 in bias per magnitude of threshold near LRG luminosities. Implication: this relation converts a selection-threshold drift into a bias drift; the LRG3 and BGS drifts of Appendix C.5 are derived through it, and Appendix N spells out the BGS conversion step by step.

O.4 The bright galaxy survey

Hahn et al. [15]. What was measured: the BGS target selection and its validation. Method: selection on Legacy Surveys DR9 imaging plus Survey Validation spectra. Result: BGS Bright is a magnitude-limited sample at $r < 19.5$ with redshift success above 95% across a broad range of galaxies; BGS Faint adds $19.5 < r < 20.175$ with a color selection. Implication: the fixed $r < 19.5$ limit raises the luminosity threshold with redshift, which drives the derived 10–25% bias drift across the BGS group (measured limit, derived drift; the derivation is Appendix N).

Sailer et al. [11]. What was measured: the growth of matter fluctuations at $z \lesssim 0.4$ from BGS imaging cross-correlated with Planck and ACT CMB lensing. Method: tomographic angular cross-spectra with a hybrid effective-field-theory model and per-bin bias nuisance parameters. Result: growth consistent with ΛCDM. Implication here: as with the LRG tomography, no resolution of bias structure between thin bins (measured context).

O.5 The emission-line galaxies

Raichoor et al. [17]. What was measured: the ELG target selection, $0.6 < z < 1.6$, and its spectroscopic efficiency. Method: a g-band fiber magnitude cut plus a $(g-r)$ versus $(r-z)$ color box; redshift reliability is a cut on the signal-to-noise of the [OII] 3727 Å doublet. Result: the [OII] signal-to-noise drops at the specific redshifts where the doublet lands on sky emission lines, especially at $z > 1.5$, and the catastrophic-failure fraction rises over $1.3 < z < 1.6$.

Implication: these windows, each $\Delta z \approx 0.001$–0.01 wide, are the one short-separation selection mechanism found in any tracer; their bias amplitude is assumed, of order 1–2%, a judgment from the weak dependence of ELG bias on [OII] flux rather than a measured bound.

Rocher et al. [18]. What was measured: the HOD of the DESI One-Percent ELG sample. Method: fits of AbacusSummit simulation halos to the measured small-scale clustering, in two redshift bins, $0.8 < z < 1.1$ and $1.1 < z < 1.6$. Result: every HOD parameter agrees within one sigma between the bins; the mean host halo mass is $\log_{10} M_h = 11.78^{+0.03}_{-0.04}$ at mean redshift 0.95 and 11.86 ± 0.05 at 1.325. Implication: the ELG bias shape is stable across ELG1 and most of ELG2 (measured); at these host masses the halo bias depends weakly on mass, so the implied drift is at the percent level (derived, with the slope assumed).

P How the prior enters the Fisher matrix

A Gaussian prior adds its precision to the Fisher matrix:

$$F \;=\; F_{\rm data} \;+\; F_{\rm priors}, \tag{81}$$

where $F_{\rm data}$ is the usual data Fisher matrix and $F_{\rm priors}$ holds one block per prior. The projected precision P of each group is added to the rows and columns of that group's bin amplitudes β_i. All other parameters (the cosmological parameters, and the velocity dispersions with their own independent Gaussian priors) are unchanged: P has no entries outside the bias block.

One consequence matters in practice. Bins that lost all their band powers to the scale cut carry no data, so their rows of $F_{\rm data}$ are zero. In the independent model such a parameter is completely unconstrained and must be dropped from the matrix before inversion. With the correlated prior it stays: P ties the empty bin to its neighbors, which do carry data, so the matrix remains invertible and the empty bin follows the chain.

Q Numerical verification

The companion script `verify_prior.py`, in this folder, checks every identity used above: the two-bin closed forms of Eqs. (33) and (36); the three-bin inverse of Eq. (52), including its exactly zero $(1,3)$ entry; the null direction, positive semidefiniteness and rank $n-1$ of P on uneven redshifts; the tridiagonal inverses of Eqs. (56) and (58); the Pythagorean identity of Eq. (44) on random displacements; the equality of P with the Schur complement of the joint (β, B) precision of Eq. (49); the decorrelation and variance of the neighbor-prediction leftover of Appendix K; the jump variance of Eq. (10) by Monte Carlo (10^5 draws); the worked example of Appendix B.2; the closed-form bias evolution of Eq. (15) against a direct integration of the continuity equation; and the tilt and neighbor-change tables of Appendix C. For the memory model of Appendix E it checks, in the same way: the annihilation $\mathsf{P}_{\rm mem}\mathsf{U} = 0$ with rank $n-2$ on uneven redshifts; the $1/W^2$ approach of the Woodbury inverse of Eq. (68) to the projection; the null-space identity of Eq. (69); the three-bin curvature penalty with its closed-form variance $2\sigma_b^2(1-r)(3-r)$; the vanishing of the projection for one and two bins; the universal-kriging predictor of Eq. (22) against a direct Gaussian conditional; the end-to-end correlation of Eq. (21) tending to ± 1; the two straight-line fits of Appendix E.4, the conserved-population table and the BGS flux-limit drift; and the LRG3 magnitude-limit rate,

0.27 magnitudes per $\Delta z = 0.1$ at $z \approx 1$ with its 6–11% bias drift. Each check prints a PASS or FAIL line with its measured deviation.

Every figure of the note lives in its own folder under `plots/`, named after the figure. The folder holds the figure's data (`data.json`), one Python script that draws it, and the PDF the note includes, so each figure redraws from its folder alone and a plot can be adjusted for the paper without rerunning the forecast. The script `plots/make_theory_data.py` writes the theory data, from the same flat ΛCDM growth factor and project fiducial amplitudes as the checks above; the notebook `EXAMPLE_EVALUATE_FISHER.ipynb` exports the FoM data.

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